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Research article

Root-mediated hydraulic performance of fibre-reinforced vegetation concrete (VC): The influence of root architecture, spatiotemporal evolution, and root vitality state

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ABSTRACT

The hydraulic performance of vegetation concrete is critical for ecological slope protection; its regulation within this engineered substrate—particularly the coupled effects of fibre incorporation, root architecture and vitality dynamics—remains unclear. This study investigated **palm fibre** and **three herbs** (*Cynodon dactylon*, *Lolium perenne*, *Paspalum notatum*) for regulating its hydraulic performance. Under varying **growth durations (30, 60, 90 d)** and **root vitality states (viable vs. devitalised)**, we analysed **root morphology, physical and hydraulic properties, soil-column dynamic infiltration, and Pearson correlations** to reveal the regulatory mechanisms of fibre-root-substrate interactions.

Results: Palm fibre reconstructed vegetation concrete's pore structure by connecting and dividing pores, forming a three-dimensional network. It enhanced water holding capacity, delayed infiltration, reduced bulk density, and increased pore diversity to facilitate root growth. Fibrous roots (*Cynodon dactylon*) with palm fibre formed dense micropores, enhancing water retention and inhibiting seepage; vertical taproots (*Lolium perenne*) tended to form preferential flow, alleviated by palm fibre's physical interception; coarse roots (*Paspalum notatum*) with palm fibre built stable composite skeletons, balancing water retention and long-term stability. Palm fibre promoted root development by physical-biochemical coupling with roots and matrix, synergising with root exudates and microbes to cement and stabilize pores, enhancing water holding capacity. Devitalised roots ceased physiological functions and lost biological cement sources, causing water-holding pore collapse; rotten root pores formed new preferential paths, leading to hydraulic deterioration (root-type dependent). Residual palm fibre network effectively delayed this process. This study provides a theoretical basis for the regulation of hydraulic performance of vegetation concrete in slope ecological protection.

1. Introduction

Plant roots improve soil performance through both physical binding and biochemical mechanisms (Wang et al., 2014) (Wang and Zhang, 2017) (Adamczyk et al., 2019). They also strongly influence soil hydraulic performance (Ng et al., 2015). Physically, roots anchor soil particles. This improves soil aggregation and pore structure, which modifies hydraulic performance (Liu et al., 2022). Conversely, decomposing roots form macropores as their biomass, diameter, and length reduce. This loosens the soil structure. These changes alter soil architecture and enhance water infiltration (Meng et al., 2025). Biochemical interactions between roots and soil also profoundly affect hydraulic

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performance. For instance, soil microorganisms like arbuscular mycorrhizal fungi (AMF) colonise plant roots. This expands the roots absorption area and strengthens water uptake capacity. Furthermore, roots absorb water for transpiration, which reduces soil moisture content and permeability whilst increasing suction (Bertolini et al., 2023). Additionally, cementing substances from soil and rhizosphere microorganisms promote soil aggregation, optimize pore structure, and improve hydraulic performance (Parhizkar et al., 2020). Intrinsic root parameters—including length, diameter, density, and distribution—also directly regulate hydraulic performance (Zi et al., 2023) (Yu et al., 2013). Ma et al. demonstrated that "the root effects of mixed plants on soil hydrology were stronger" (Ma et al., 2024). They identified an optimal root density for hydraulic performance, based on saturated hydraulic conductivity and moisture content. In summary, research on root-mediated soil hydraulic performance is now a mature and well-established field.

Technological advances now allow plant roots to interact with novel engineering materials, not just natural soils. In civil engineering, bare slopes pose major geotechnical hazards. These include severe erosion, structural instability, and substantial soil and water loss. Ecological protection strategies therefore advocate using plant roots to reinforce slopes. However, field evidence shows that natural revegetation is often prohibitively slow. Consequently, various artificial ecological slope-protection technologies have been developed. These techniques accelerate vegetation recovery and reinforce the slope surface, leading to widespread global adoption. The core of these systems is the slope-protection substrate. One key material is Vegetation Concrete (VC). VC is composed mainly of planting soil, cemented with cement, and supplemented with organic materials, fertiliser, and water. It effectively protects slopes from 45° to 75°(Liu et al., 2023). VC ecological restoration technology has now been applied in over 20 Chinese provinces, covering more than 35 million square metres. It has established industry standards and is gaining global reach. This comprehensive development demands continued research into VC. Current research focuses on how the environment affects substrate performance, on improving the substrate mix, and on evaluating VC's ecological efficacy.

However, interactions between plant roots and the VC substrate remain poorly understood. Although valuable insights exist—for example, Xiong et al. found that roots improve VC's anti-erosion performance and shear strength, and Xia et al. reported that Vegetation alters the microbial community and functional diversity of VC—substantial knowledge gaps remain in this critical area (Xiong et al., 2023) (Xia et al., 2022).

The vegetation concrete (VC) ecological slope restoration project along the Tian'e-Beihai Highway (Nanning to Pingguo, Guangxi, China) faces a key challenge: expansive soils (Fig. 1). These soils are highly water-sensitive. Their clay minerals swell and shrink with moisture changes, causing substantial volumetric shifts that lead to slope cracking and instability. In this setting, plant roots have a dual role. They anchor soil and reinforce the slope, but they also modify the substrate's pore structure, which can enhance water infiltration. Understanding root-mediated water transport in VC is therefore crucial. In practice, palm fibre (PF) is routinely incorporated to suppress this shrinkage-swelling damage (Shen et al., 2021) (Wang et al., 2023a) (Wang et al., 2023b). This material is selected to meet the nutrient needs of plants in ecological restoration. However, adding fibre also affects VC's hydraulic performance by altering its physical structure. The fibres also alter the chemical environment, creating favorable conditions for microorganisms. This promotes microbial respiration, arbuscular mycorrhizal fungi (AMF) colonization, and subsequent root development. Furthermore, cementing materials—including root exudates and mycelial networks—bind soil particles and form micropores. Additionally, the fibre network provides a scaffold for microbial colonization and pore interconnection, further modifying hydraulic performance. In summary, the combined effects of plant roots and palm fibre on VC's water transport are not well understood. Existing research also focuses predominantly

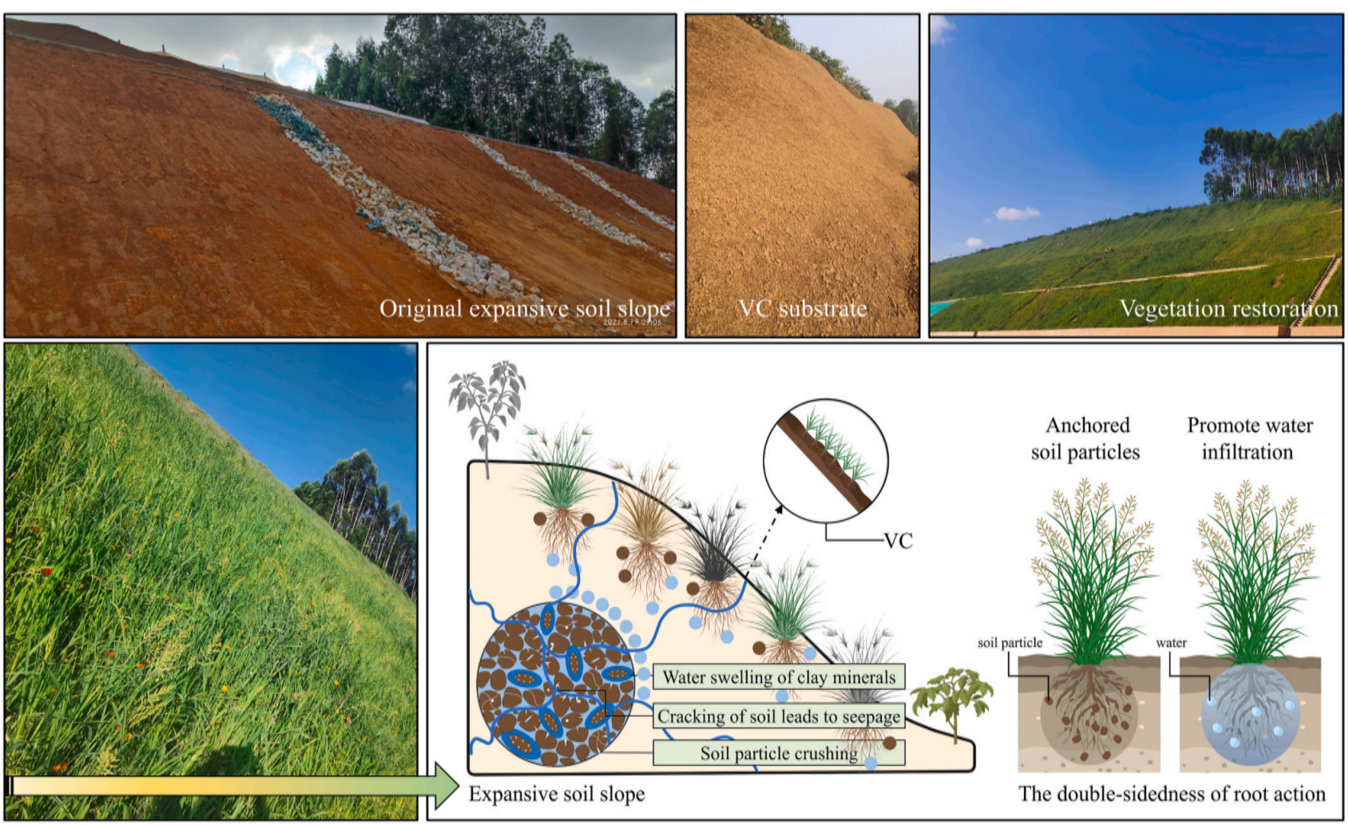

Fig. 1. Role of plant root systems in Vegetation Concrete (VC) slope restoration: Tian'e–Beihai expressway (Nanning to Pingguo Section), Guangxi, China.

on single permeability metrics. Therefore, experimental research on how palm fibre and roots affect VC's water transport is essential.

Notably, current understanding of root-mediated regulation of hydraulic performance is predominantly based on natural soil systems. However, the mechanisms controlling hydraulics within artificial composite substrates like Vegetation Concrete (VC) remain poorly understood. Specifically, research on VC exhibits several limitations. Firstly, the research focus is unbalanced. Existing work has largely concentrated on how roots improve the physical-mechanical properties and biochemical characteristics of VC, while systematic investigation into its hydraulic performance is notably lacking. Secondly, material synergy is overlooked. Research on palm fibre (PF), commonly incorporated in engineering, has similarly emphasised mechanical enhancement. Little is known about how PF influences the hydraulic properties of VC, or how it interacts synergistically with plant roots to regulate moisture. Most critically, the prevailing research perspective is static and oversimplified. Most studies have examined only a single plant species or a fixed growth stage. They have neither systematically compared the differential effects of distinct root morphologies, nor tracked the evolution of hydraulic performance throughout plant growth and the critical transition of roots from a viable to a devitalised state. This study addresses these gaps by designing a comprehensive controlled experiment. It concurrently considers PF incorporation, three typical root morphologies, and the spatiotemporal evolution of root vitality (Dimitriadis et al., 2024b). The aim is to systematically reveal the hydraulic interaction mechanisms within the root-fibre-substrate complex under multifactorial coupling.

To achieve this aim, this study investigated vegetation concrete (VC) and palm fibre-modified VC (PF–VC). For initial slope protection with cement-based habitat substrates, plant establishment follows two principles: using pioneer species and a herb-dominated species mix. We therefore selected three herbaceous pioneer species with distinct root morphologies (*Cynodon dactylon*, *Lolium perenne*, and *Paspalum notatum*) for an outdoor pot experiment. The experiment tested the effects of growth duration (30, 60, and 90 days) and root vitality (viable roots, VR; devitalised roots, DR). The detailed experimental design is provided in Table S1. This study systematically explored how the incorporation of palm fibre (PF), in synergy with plant roots, alters the hydraulic performance of VC. It addressed three key questions: (1) the mechanisms—both direct and indirect—by which PF incorporation regulates the hydraulic performance of VC; (2) how plant root morphology and vitality alter the hydraulic performance of the root-fibre-VC complex; and (3) whether the synergy between PF and plant roots can mitigate hydraulic deterioration caused by root devitalisation, and the underlying regulatory mechanisms. This work provides a theoretical basis for regulating VC's hydraulic performance in slope protection. It supports the optimal design and application of VC, potentially reducing long-term maintenance costs and enhancing project durability (Dimitriadis et al., 2024a) (Dimitriadis et al., 2025).

2. Materials and methods

2.1. Materials

The materials comprised expansive soil (ES), standard soil, cement, habitat material modifying agent (HMMA), organic fertilizer (OF), habitat substrate organic material (HSOM), PF, and water. ES was sourced from the Guangxi Ping nan Expressway project site. It was then dried, crushed, and sieved through an 8 mm mesh to remove plant rhizomes, stones, and oversized particles. Table S2 details the basic physical properties of the sieved ES, while Fig. 2 illustrates its sampling process, particle size distribution, and mineral composition. The standard soil was a sandy loam from a greening project in Yichang City, Hubei Province. It underwent the same processing as the ES; Its fundamental physical properties are listed in Table S3. We used P·O 42.5 ordinary Portland cement (Yichang Huaxin Cement Co., Ltd., China). HMMA (Hubei Runzhi Ecological Technology Co., Ltd., China) was included to enhance substrate physicochemical properties, adjust biological characteristics, modify microstructure, and promote plant growth. The OF (Hubei Chubang Ecological Agriculture Co., Ltd., China) was a selenium-enriched type. The HSOM (Yichang Jinzheng Rice Industry Co., Ltd., China) was prepared from sieved, dried rice husk (RH). PF was sourced from Shandong Guangsu Agricultural Technology Co., Ltd. as consolidated mats, which were then processed into discrete 12 mm-long fibres. Its properties are summarised in Table S4. Photographs of all materials appear in Fig. S1.

2.2. Experimental design and index determination

2.2.1. Pot experiment

We established 72 flowerpots (23 cm diameter × 21.5 cm height) in triplicate for each curing age, including blank controls. Each pot was filled with stratified layers: 10 cm of VC over 10 cm of ES. The bulk density was controlled at 1.3 g cm^{-3} for VC and 1.49 g cm^{-3} for ES. The mass of components for each pot was calculated from the layer volumes. The VC mix proportions are given in Table S5. The PF content was 0.4 % of the dry mass (NB/T 35082, 2016) (Liu et al., 2025b). We then added deionised water (20 % of the dry mass) and mixed thoroughly. The mixture was compacted into the flowerpots in uniform layers, and the pots were labelled. The substrate was then gradually saturated with

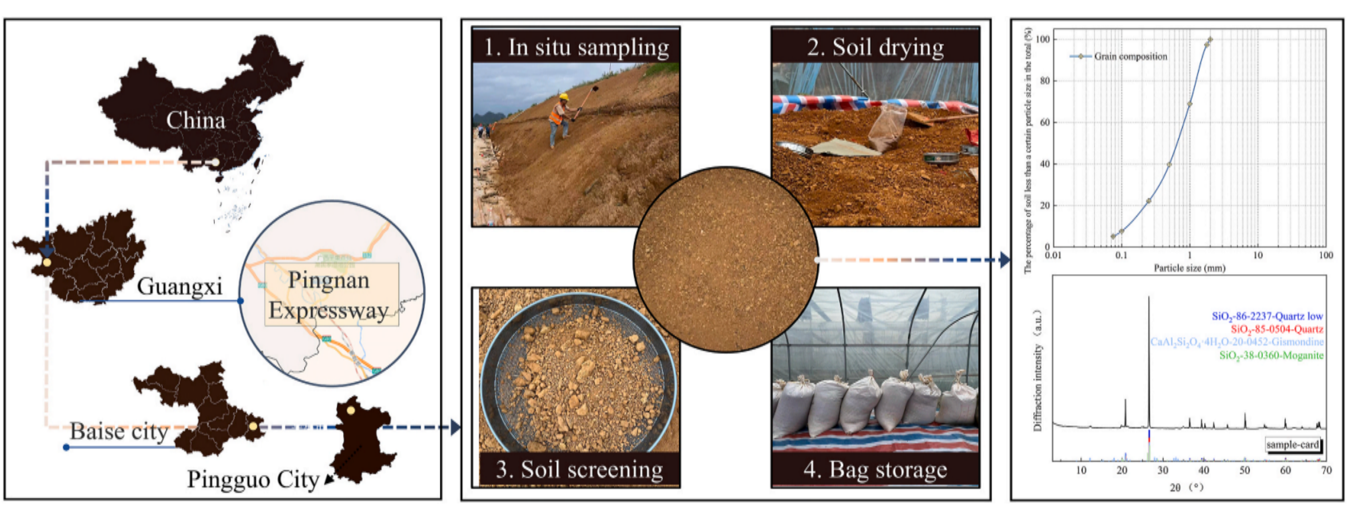

Fig. 2. The preparation process, particle size distribution and XRD test results of ES raw materials.

water. Seeds (0.98 g per pot) were sown at a density of 20 g m^{-2}. Seed morphology is shown in Fig. S2. A fine soil cover (2–4 mm) was applied and remoistened. All pots were placed outdoors and watered on a scheduled regime. At designated ages, the shoots were cut from the root devitalisation groups. The pots were sealed in lightproof, waterproof bags to simulate three drought-induced conditions: (i) hydraulic deprivation; (ii) photosynthetic blockade; and (iii) root system devitalisation. Sampling at each curing age included:

SWCC testing: Three cutting ring samples (61.8 mm diameter × 20 mm height) were taken vertically from different depths in six specimen groups per pot;

Variable-head permeability testing: Additional rings (61.8 mm diameter × 40 mm height) were collected for this test;

Soil-column permeability testing: Intact columns (20 cm high × 10 cm diameter) were obtained.

Root systems were also collected and preserved at the same time. Fig. 3 illustrates the complete testing and sampling procedure.

2.2.2. Physical property index test

Soil tests followed the Chinese national standard GB/T 50,123-2019. Cutting ring samples were extracted from the flowerpot for analysis. Rings measuring 61.8 mm in diameter × 20 mm in height were used for bulk density (BD), while rings of 61.8 mm × 40 mm were used for total porosity (TP) and saturated moisture content (SMC). Three parallel samples were prepared for each test. The testing procedures were as follows:

BD, TP and SMC were determined by the saturated drainage method. The procedure was:

1. Weigh a dry cutting ring of known volume, V (cm^3), as m_1 (g).
2. Soak the sample in water for 24 h and weigh the saturated mass, m_2 (g).
3. Place the sample in naturally air-dried coarse sand for 6 h, then weigh after natural water loss, m_3 (g).
4. Oven-dry the sample at 105 °C for 24 h to constant weight and weigh, m_4 (g).

The density of water, ρ_w, was taken as 1.00 g cm^{-3}. The parameters were calculated as follows:

$$BD = \frac{m_4\text{-}m_1}{V} \tag{1}$$

$$TP = \frac{m_2\text{-}m_4}{V\cdot\rho_w} \tag{2}$$

$$CP\ (Capillary\ Porosity) = \frac{m_3\text{-}m_4}{V\cdot\rho_w} \tag{3}$$

$$NCP\ (Non\text{-}capillary\ Porosity) = TP\text{-}CP \tag{4}$$

$$GWR\ (Gas\text{-}Water\ Ratio) = \frac{NCP}{CP} \tag{5}$$

$$SMC = \frac{m_2\text{-}m_4}{m_4} \tag{6}$$

2.2.3. Saturated permeability coefficient (SPC) test

The saturated permeability coefficient (SPC) was measured using a TST-55 permeameter with the variable-head method. Samples were held in flanged cutting ring (61.8 mm diameter × 40 mm height), with three parallel replicates prepared. The permeability coefficient was calculated as follows:

$$k_t = 2.3\frac{A_n L}{A(t_2\text{-}t_1)} lg\frac{h_1}{h_2} \tag{7}$$

The terms in the formula are defined as follows: k_t is the permeability coefficient (cm·s^{-1}) at water temperature t° C; A_n is the cross-sectional area of the graduated glass tube (cm^2); A is the cross-sectional area of the sample (cm^2); L is the sample height (cm); t_1 and t_2 are the start time and end time of the test (s); and h_1 with h_2 are the initial and final water

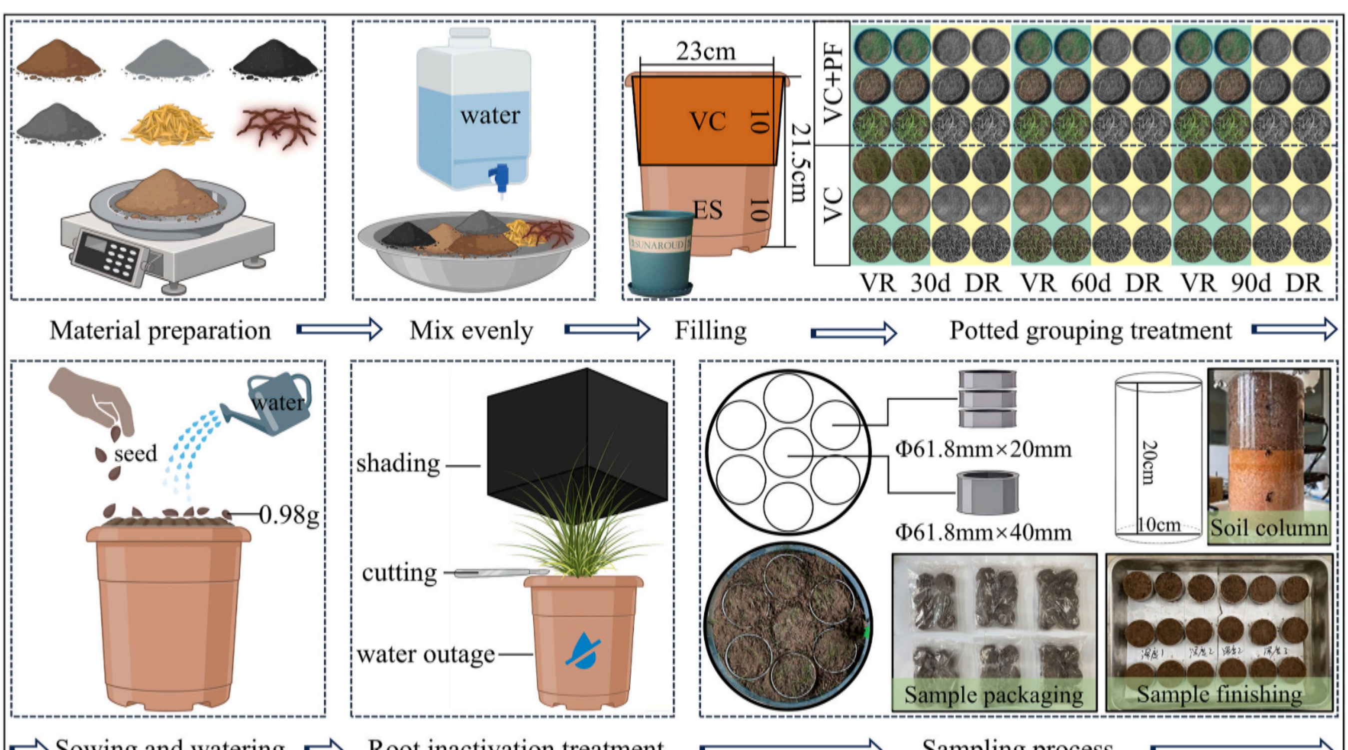

Fig. 3. Pot experiment and sampling process.

heads, respectively (cm). Each sample underwent five variable-head permeability tests, and the average coefficient was reported.

2.2.4. Soil-water characteristic curve (SWCC)test

The SWCC is a fundamental relation in unsaturated soil mechanics. It describes the relationship between matrix suction and soil moisture content, effectively reflecting the soil's water holding capacity (Natalia and Yang, 2025). Matrix suction was measured using the filter paper method (Muhammad and Siddiqua, 2021). As Fig. 4 shows, two large protective filter papers and one small test filter paper were sandwiched between two cutting rings in a large-small-large sequence. We used Whatman No. 203 slow filter papers (Cytiva). The protective layers were qualitative filter paper discs (61.8 mm diameter), and the test layer was a quantitative filter paper disc (50 mm diameter). Before use, the filter papers were oven-dried and stored in desiccant-filled aluminum boxes. Filter paper moisture content was determined using an analytical balance with 0.0001 g accuracy. After placeing the filter papers, the samples were immediately sealed with electrical tape. The assembly was then wrapped in multiple layers of cling film and stored in a sealed tank. The assembly equilibrated for at least 7 days to ensure sufficient moisture exchange between the soil and filter papers. The test filter paper was then weighed to determine its moisture content, and the matrix suction was calculated using the calibration equation:

$$\begin{cases} lg\,S = 5.493 - 0.0767\omega_{fp}, \omega_{fp} \leq 47\% \\ lg\,S = 2.470 - 0.012\omega_{fp}, \omega_{fp} > 47\% \end{cases} \tag{8}$$

where S and ω_{fp} are matrix suction (kPa) and filter paper moisture content (%), respectively.

After the filter paper test, the soil sample was weighed to obtain its wet mass, then oven-dried at 105 °C for 24 h to determine its dry mass. The mass moisture content was calculated for each sample, and the volumetric moisture content was derived by multiplying it by the

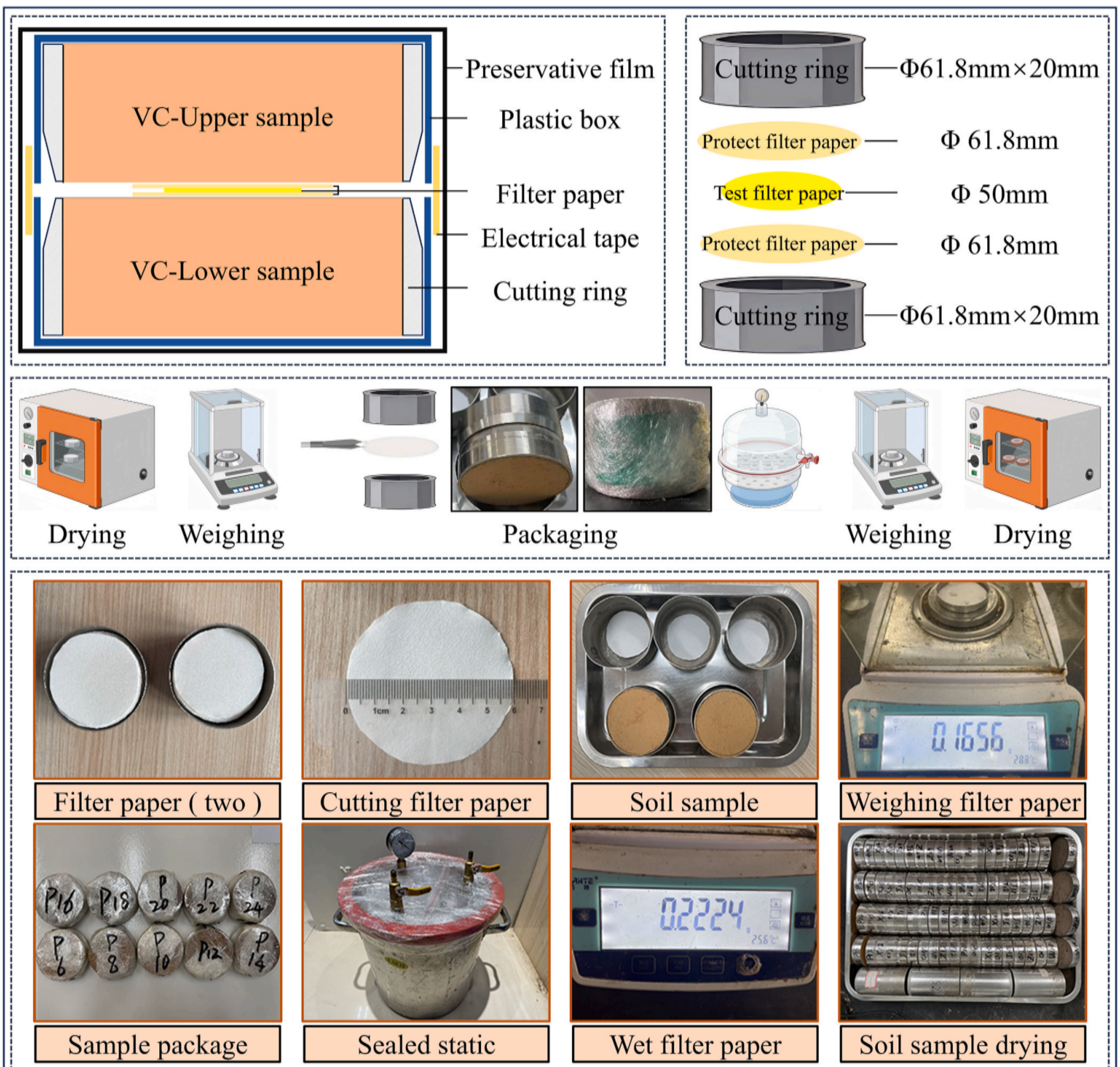

Fig. 4. The schematic diagram of matrix suction measured by filter paper method; Soil water characteristic curve test flow chart.

sample's volumetric weight. The complete testing procedure is outlined in Fig. 4.

The SWCC was fitted using the van Genuchten (VG) model (Ghorbani et al., 2025):

$$\frac{\theta(h)\text{-}\theta_r}{\theta_s\text{-}\theta_r} = \left[\frac{1}{1+(\alpha h)^n}\right]^m \tag{9}$$

In the formula, $\theta(h)$ represents the volumetric water content ($cm^3 \cdot cm^{-3}$); θ_s is the saturated volumetric water content ($cm^3 \cdot cm^{-3}$); and θ_r is the residual volumetric water content ($cm^3 \cdot cm^{-3}$). The term h denotes the matrix suction (kPa); The parameter α relates to the air-entry value and is positively correlated with its reciprocal; m and n are shape coefficients, $= 1\text{-}\frac{1}{n}$, n is the slope of the corresponding point of the air-entry value, which is related to the pore size distribution of the medium.

2.2.5. Root scanning test

Following the permeability and SWCC test, soil samples were immersed in deionised water. After the samples disintegrated and settled, the floating roots were carefully extracted with tweezers and rinsed with distilled water for scanning. Root images were acquired using an Epson Perfection V700 Photo root scanner. These images were then analysed with the WinRHIZO system to determine root diameter, length, surface area, and volume distributions. Root parameter distributions are presented as box plots. Scanning electron microscopy (SEM) images were obtained using a field-emission instrument; further equipment details are provided in Fig. S3.

2.2.6. Constant water head soil column permeability test

We conducted infiltration experiments using a vertical one-dimensional infiltration apparatus. The setup included a soil column and a Mariotte bottle (Fig. 5). The soil column (10 cm diameter, 20 cm height) had a 5 mm precision scale engraved on its surface. Its 10 cm-high base contained a perforated plate for air release and stability. Pre-drilled ports for three-needle moisture probes were spaced at 5 cm intervals along the side. A detachable rainfall simulator (11 cm high) was mounted on top. During testing, the Mariotte bottle maintained a constant 3 cm water head in the upper chamber. Additional soil columns with bayonet-fit bases were used for field sampling. Their lower edges were sharpened to aid specimen collection. After collection, moisture content was measured using time-domain reflectometry (TDR) with a VMS-3000-TR soil sensor (Wang et al., 2025a). Before starting, moisture

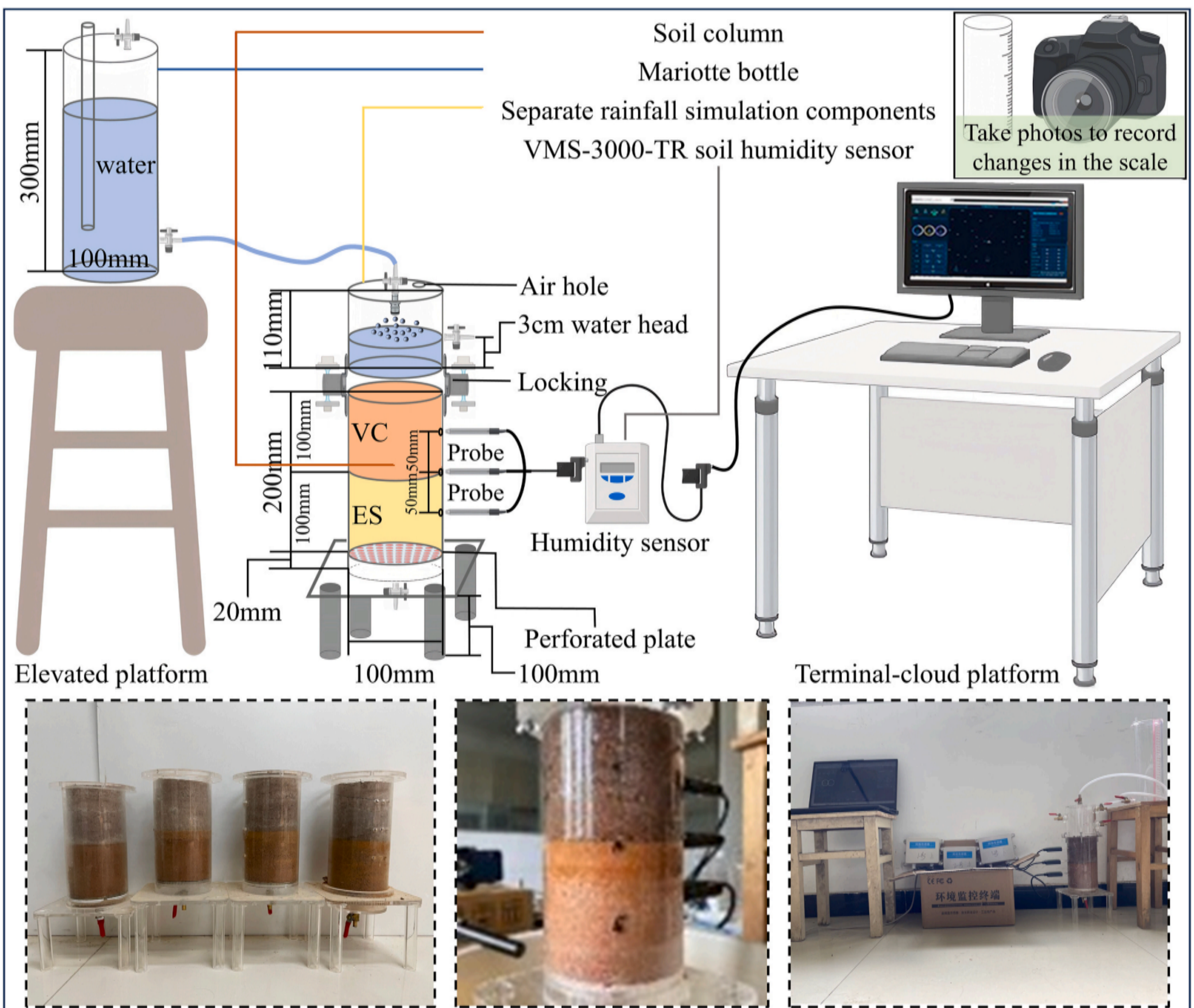

Fig. 5. Constant water head soil column permeability test device and test schematic diagram.

probes were inserted to specified depths, with data recorded at 1 min intervals. The test began only after stable readings were confirmed. The system automatically recorded and transmitted the mass moisture content of each soil layer to the terminal.

2.3. Statistical analysis and visualization

Parallel test data are reported as the mean ± standard error from three replicates. Data were analysed statistically using IBM SPSS Statistics 25 and Microsoft Excel 2016. Significance and correlations were assessed by one-way analysis of variance (ANOVA) and Pearson's correlation analysis. Images were processing and figures were generated using Adobe Photoshop 2024, Origin Pro 2022, and Microsoft Office PowerPoint. The RETC software (USSL) was used to fit the experimental relationship between matrix suction and volumetric water content.

3. Results and analysis

3.1. Root scanning

3.1.1. Root morphology

The complete root scanning images of the three plants across each growth duration were divided into PF–VC and VC groups, as shown in Fig. 6. Plant roots rotted and decomposed after devitalisation treatment, meaning that root parameters of the devitalised group could not be collected. The shaded part represents the planar morphological characteristics of the root system, and its area corresponds to the root projection area. Fig. 6 shows that root morphological characteristics were broadly similar between the PF–VC and VC groups for the same plant at the same growth duration. Regarding root length, tillering, and opening and closing, growth and development were largely consistent across groups at identical growth durations. With prolonged growth duration, roots of all plants gradually enlarged, and tiller number increased, but distinct differences emerged in root architecture and spatial distribution. Specifically, Plant (*Cynodon dactylon*) G's roots developed continuously from the main root, forming new roots and establishing a relatively dense, shallow root system. At 30 d and 60 d growth durations, its roots were primarily distributed in the 0–2 cm soil layer; by 90 d, they occupied the 0–4 cm layer. Plant (*Lolium perenne*) H exhibited a straight root development model, with its main root elongating over extended growth durations. Its root system had significantly fewer tillers than that of Plant G, though at the 30 d growth duration, its developmental progression closely resembled that of Plant G's root system. Plant (*Paspalum notatum*) B developed a relatively extensive tillering root system at the 30 d growth duration. Initially, roots grew upright, later exhibiting an inclined growth angle. In root penetration depth, it showed high similarity to Plant H, roots distributed at 0–2 cm (30 d), 0–4 cm (60 d), and 0–6 cm (90 d).

Table 1 presents the. The PF–VC group exhibited consistently slightly larger root projection areas across all growth durations, indicating that PF incorporation favors plant root development. This occurs because PF incorporation enhances substrate pore structure during early growth stages, improving moisture retention and creating more favorable conditions for plant growth (Wang et al., 2021). Concurrently, the natural degradation of PF in soil supplies organic nutrients to the substrate, thereby increasing nutrient availability for plants (Fode et al., 2024) Such conditions hinder root growth within the soil matrix. Furthermore, root projection areas for all three plants in both PF–VC and VC groups followed the hierarchy Plant H > Plant B > Plant G at every growth duration. Based on root projection area measurements, the root system of H extended most rapidly, that of Plant B exhibited intermediate growth speed, and Plant G progressed slowest.

3.1.2. Morphological parameter

The total root length, ground diameter, root surface and root volume of Plant G, Plant H and Plant B in the PF–VC and VC groups at 30 d, 60 d and 90 d were presented in Table 2. According to the table and significance analysis, the root parameters of different plants differed extremely significantly different ($P < 0.01$). Simultaneously, the root parameters of each plant at different growth durations also showed extremely significant differences ($P < 0.01$). Additionally, extremely significant differences ($P < 0.01$) were observed in root parameters between corresponding samples of the PF–VC and VC groups, indicating that PF incorporation had a greater impact.

Additionally, the box plot was used to visually present the distribution of parameters including total root length, ground diameter, root surface and root volume within each experimental group. This approach aims to minimize potential systematic errors introduced during sampling and scanning. Fig. 7 shows that all root parameters were significantly larger in the PF–VC group than in the VC group across the periods except for the total root length of Plant G at 90 d, where the VC group value was slightly larger. This indicates that fibre incorporation promotes root development in all plants. When comparing root parameters across growth durations, the relative magnitudes of total root length,

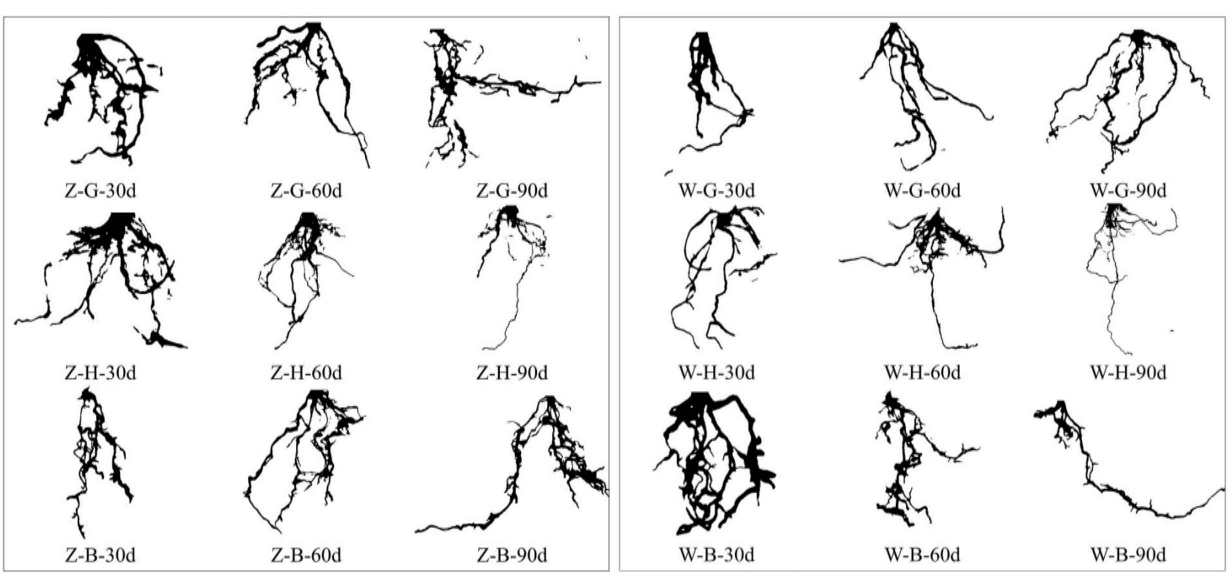

Fig. 6. Root scanning images of three plants at each growth duration. Z and W represent PF–VC group and VC group, respectively. *Cynodon dactylon* (G), *Lolium perenne* (H), *Paspalum notatum* (B). 30 d, 60 d, 90 d were three growth duration.

Table 1
Plant root projection area. (cm^2).

Test groups	G			H			B		
	30 d	60 d	90 d	30 d	60 d	90 d	30 d	60 d	90 d
PF – VC	0.61	0.81	1.01	1.22	2.43	3.60	0.84	1.22	1.95
VC	0.43	0.65	0.92	0.75	1.72	2.21	0.63	0.87	1.16

Note: *Cynodon dactylon* (G), *Lolium perenne* (H), *Paspalum notatum* (B). 30 d, 60 d and 90 d were the three growth durations.

Table 2
Root parameters of three plants in each sample group at different growth durations.

Test groups	Growth duration/d	Vegetation type	Total root length/cm	Ground diameter/mm	Root surface/cm^2	Root volume/cm^3
PF–VC	30 d	G	12.58 ± 2.41Cb	0.53 ± 0.03Cc	2.17 ± 0.32Cc	0.032 ± 0.005Ac
		H	18.45 ± 1.87Ca	0.67 ± 0.04Ca	3.77 ± 0.12Ca	0.062 ± 0.002Ca
		B	13.38 ± 2.24Cb	0.65 ± 0.09Cb	2.44 ± 0.1Cb	0.047 ± 0.005Bb
	60 d	G	16.11 ± 0.77Bb	0.55 ± 0.02Bc	2.50 ± 0.08Bc	0.030 ± 0.005Ac
		H	31.75 ± 1.02Ba	0.71 ± 0.02Bb	7.59 ± 0.17Ba	0.136 ± 0.002Ba
		B	16.90 ± 3.41Bb	0.72 ± 0.05Ba	4.24 ± 0.9Bb	0.064 ± 0.017Bb
	90 d	G	17.94 ± 0.84Ac	0.58 ± 0.02Ab	3.13 ± 0.09Ac	0.032 ± 0.005Ac
		H	38.58 ± 1.55Aa	0.78 ± 0.01Aa	12.60 ± 0.18Aa	0.247 ± 0.045Aa
		B	30.25 ± 1.95Ab	0.82 ± 0.10Aa	6.71 ± 1.36Ab	0.142 ± 0.038Ab
VC	30 d	G	7.61 ± 1.41Cc	0.39 ± 0.02Cc	0.98 ± 0.14Cc	0.01 ± 0.003Cc
		H	13.05 ± 1.98Ca	0.53 ± 0.03Cb	2.26 ± 0.36Ca	0.03 ± 0.006Ca
		B	9.87 ± 2.23Cb	0.56 ± 0.04Ca	1.71 ± 0.14Cb	0.03 ± 0.003Cb
	60 d	G	12.99 ± 1.95Bb	0.43 ± 0.02Bc	1.52 ± 0.26Bc	0.02 ± 0.002Bc
		H	24.29 ± 2.72Ba	0.56 ± 0.02Bb	5.46 ± 0.17Ba	0.08 ± 0.007Ba
		B	13.00 ± 1.65Bb	0.62 ± 0.02Ba	1.96 ± 0.19Bb	0.03 ± 0.002Bb
	90 d	G	20.66 ± 1.95Ab	0.46 ± 0.02Ac	2.21 ± 0.15Ac	0.03 ± 0.004Ac
		H	34.85 ± 4.29Aa	0.66 ± 0.03Ab	7.62 ± 0.99Aa	0.12 ± 0.025Aa
		B	18.61 ± 2.07Ab	0.71 ± 0.03Aa	2.99 ± 0.38Ab	0.05 ± 0.009Ab

Note: A-C, Different letters indicate significant differences ($p < 0.05$) in root parameters at different growth durations. a-c, Different letters indicate significant differences ($p < 0.05$) in root parameters among different plant types.

surface, and volume followed Plant H > Plant B > Plant G in both treatment groups at all durations. In contrast, the order for root diameter was consistently Plant B > Plant H > Plant G. With increasing growth duration, all parameters showed an overall increasing trend, with these relative magnitude trends becoming more pronounced. Plant H exhibited the fastest growth rate, followed by Plant B, and Plant G the slowest.

Among the parameters, total root length and ground diameter determine root surface and volume (Zhang et al., 2024). Moreover, changes in root surface area and volume mirrored those in total root length trends. Therefore, the analysis focuses on total root length and ground diameter. In the PF–VC group, the total root length of Plant H and Plant B increased by 46.6 % and 6.4 % at 30 d, 97.1 % and 4.9 % at 60 d, and 115.1 % and 68.6 % at 90 d, respectively, compared to Plant G. This indicates that the gap between the total root length of Plant H and Plant B and that of Plant G is widening, and that the root system of Plant H develops faster than that of Plant B. The total root length increases for the three plant species were 28.1 %, 72.1 %, and 26.3 % at 60 d, respectively, and increased to 42.6 %, 109.1 %, and 126.1 % at 90 d. This indicates that as growth duration increased, the total root length of all three plant species increased significantly, in the order of Plant B > Plant H > Plant G. Plant B exhibited higher root system development in the later stage compared to the earlier stage, with a 79 % increase in total root length from 60 d to 90 d. Plant H saw approximately a doubling of total root length, while Plant G exhibited a nearly linear, gradual increase in total root length. In the VC group, the increase in total root length of Plant H relative to Plant G was 71.5 %, 87 %, and 68.7 %, respectively, maintaining a stable overall state. The increase in total root length of Plant B relative to Plant G was only 29.7 % at 30 d. As time progressed, the total root lengths of the two gradually converged, and at 90 d, the total root length of Plant G was greater than that of Plant B. The total root length increases for the three plants rose from 70.7 %, 86.1 %, and 31.7 % at 60 d to 171.5 %, 167 %, and 88.55 % at 90 d, respectively. The total root length increases for Plant G and Plant H were greater, while Plant B showed a gradual linear increase in total root length at this time.

From the box plot of ground diameter changes, it can be seen that the PF–VC group showed the greatest variation in Plant B with growth duration. At 30 d, its root diameter was 0.65 mm, slightly smaller than that of the Plant H (0.67 mm). By 60 d and 90 d, its ground diameter reached 0.72 mm and 0.82 mm, respectively, making it the plant with the largest basal diameter among the three. In contrast, the ground diameter of Plant G showed the smallest variation, increasing from 0.53 mm at 30 days to 0.58 mm at 90 d in a slow linear manner. Within the VC group, a consistent trend was observed, with Plant B having the largest ground diameter at all time points. Its ground diameter increased by 0.15 mm with age, while Plant G still had the smallest increase, of only 0.07 mm.

3.2. Physical property index

3.2.1. BD

The BD of each sample is mainly divided into four parts as shown in Fig. 8. Fig. 8a shows the change in BD of samples containing VR, and Fig. 8b shows the change in BD of samples containing DR. Since the plant roots had grown to a depth of 6 cm soil layer at 90 d, and the height of the cutting ring height used for the BD test was 20 mm, stratified sampling was performed on the 90 d specimens containing active root systems. This approach aimed to further investigate variations in BD with depth across soils containing different plant root systems, as shown in Fig. 8c. In addition, Fig. 8d shows the increment of BD of the samples with DR relative to the samples with VR. Because the difference in BD data between each sample was small, significance analysis was performed for each sample group to facilitate the analysis. The analysis results show that there was a significant difference in root activity in soil BD ($p < 0.01$). In the samples containing VR, the addition of PF had a significant effect on soil BD ($p < 0.01$). There were significant differences in BD among different root types and growing samples ($p < 0.05$). During the 90-day growth duration, there was a significant difference in

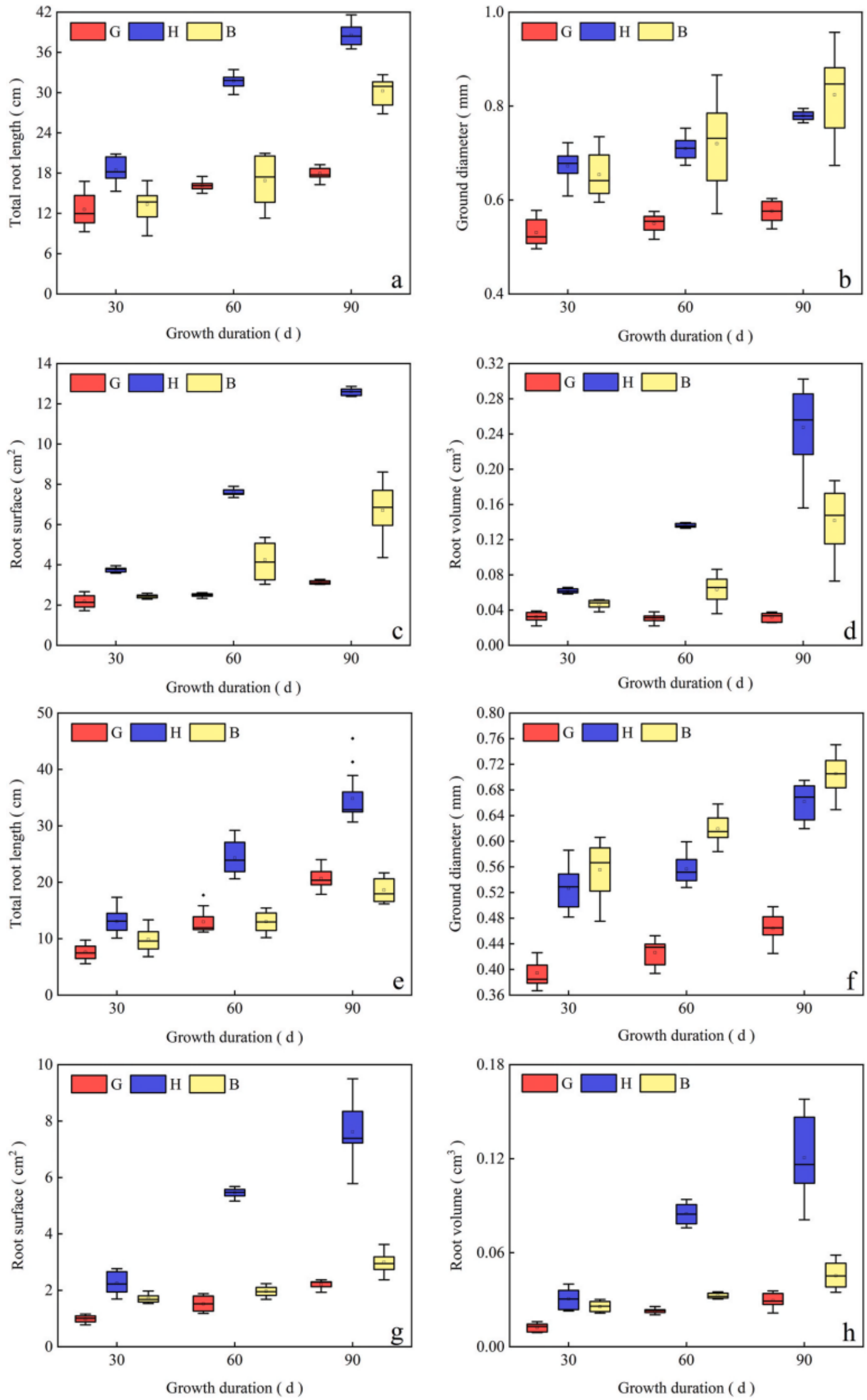

Fig. 7. The box plot of plant root parameters with growth duration in PF–VC group and VC group. a–d: PF–VC group. e–h: VC group.

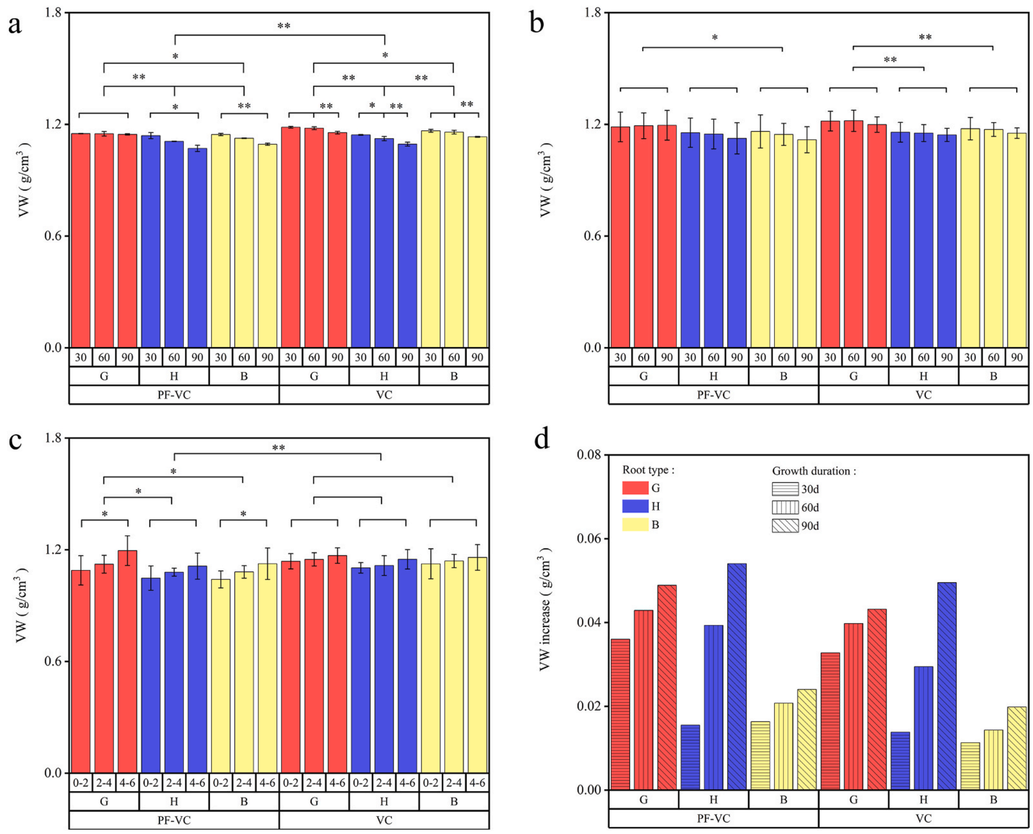

Fig. 8. The change trend of BD of each sample under different plant roots. * and ** represent significance at the 0.05 and 0.01 levels, respectively.

BD with soil depth in the PF–VC group ($p < 0.05$), but no significant difference in the VC group. Compared with the 1.244 g/cm^3 of the fibre-free and root-free control sample BD, the BD of the root-containing sample was smaller. There was no significant difference in the BD of the devitalised group at each growth duration. This shows that the effect of the root system on soil BD is obvious, and it can improve the local permeability of soil and reduce soil BD (Wang et al., 2025b).

Taking the VR sample of PF–VC group as an example, the average values of BD of root soil of different plants over three growth durations were Plant G (1.150 g/cm^3), Plant H (1.106 g/cm^3) and Plant B (1.121 g/cm^3). An inverse relationship was observed between BD and root parameters, with BD following the order Plant G > Plant B > Plant H. With the extension of growth duration, the BD of different plant root samples showed a downward trend. All showed that the decrease in the early stage (30 d–60 d) was less than that in the later stage (60 d–90 d). This change was similarly observed in the VC group. The BD of the sample with DR basically maintained the change trend of the BD of the VR sample and increased more than that of the VR sample. From the common point of view of the PF–VC group and the VC group, the plant with the largest increment was Plant H, and both increments occurred at the 90 d growth duration. It is also the plant with the largest increment amplitude from the growth duration perspective. This is attributed to the high biomass, soft and young tissue structure, and rapid decay of Plant H roots in the soil. The root channel formed by it becomes filled under natural seepage sedimentation, which in turn reduces porosity and increases BD (Tang et al., 2025). Conversely, the Plant B exhibited the smallest increase in BD, which is linked to its distinct root structure, morphology, and arrangement density.

With the increase of soil depth, the BD of root soil increased, which was related to the root density at each depth. Plant G showed the largest variation, especially in the PF–VC group. After adding PF, micro-pores increase and there are more water-retaining micropores. These micropores guided the rich fibrous roots to grow towards these more humid pores, thus compacting and binding the soil. The porosity of the surface soil increases and BD decreases. This is evidenced by the slightly larger BD of VR and DR samples in the VC group compared to the PF–VC group at each time period. Because the root penetration depth of Plant G is confined to 0–4 cm during the research period, BD also shows a steep rise at 4–6 cm.

The dense root-soil complex formed by the abundant fibrous roots of Plant G interspersing the sample surface is more conducive to soil and water conservation (Cao et al., 2025). At the same time, abundant root exudates enhance aggregate stability, collectively reducing soil compaction and deposition under natural conditions, resulting in stable pore structure and smaller BD. The root depth and diameter of Plant B and Plant H are larger, facilitating skeleton structure formation. Plant B has stout roots, rough texture, high cellulose content, slow decay rate and multi-section distribution (Ye et al., 2017). Its fibrous roots grow in close internodes, growing at an angle to the soil layer. With extended growth duration, the root system becomes more horizontal, developing more intersection points within the root zone. This inhibits soil particle sinking, stabilises soil structure, and provides longer-lasting reduction of BD due to slow decay. Plant H roots penetrate deeply, have the widest distribution roots, with root activity increasing soil porosity and reducing BD. Consequently, root distribution is negatively correlated with soil BD.

3.2.2. TP

Fig. 9 shows the change trend of porosity of different samples. In the samples containing VR, with the increase of growth duration, TP and CP of different samples showed an upward trend. Among them, TP showed Plant H > Plant G > Plant B in plant types, while CP showed differences in different growth durations. The performance of CP in three types of plants decreased slightly in turn at 30 d, and the opposite was observed at 90 d, indicating that the effect of plant root amount at different growth stages on CP may be completely opposite.

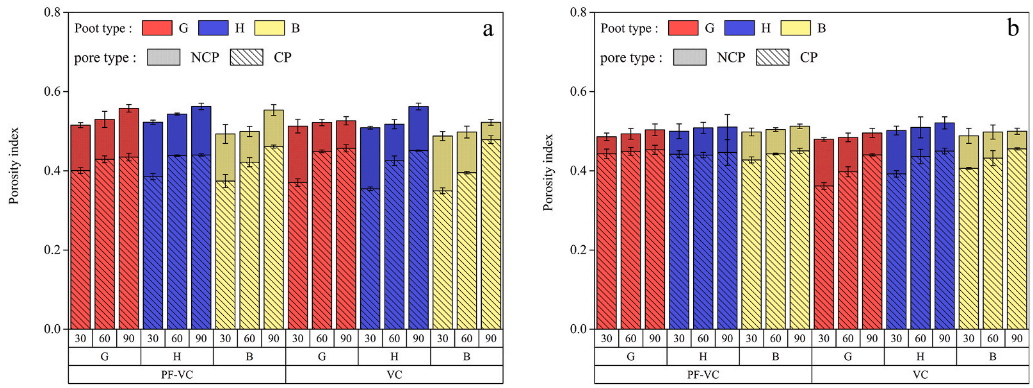

Fig. 9. The change trend of porosity of each sample under different plant roots. a: Viable root-treated sample groups. b: Devitalised root-treated sample groups.

In the samples containing VR, the average TP values of different plants in the PF–VC group were 0.534 (Plant G), 0.543 (Plant H), and 0.515 (Plant B), respectively, which were higher than 0.520, 0.530, and 0.503 in the VC group. A similar trend was observed for CP, indicating that the incorporation of PF can promote the improvement of porosity of root-containing soil. By comparing the variation range of CP in the PF–VC group and the VC group in Fig. 9a, it can be seen that the data distribution of the PF–VC group is more concentrated. By calculating the coefficient of variation (CV) as the ratio of standard deviation to mean value for both groups of porosity both, it was found that the discrete degree of TP is equivalent. In contrast, the CV value for CP in the VC group (0.117) was nearly double that of the PF–VC group (0.067). This phenomenon indicates that PF has a stabilising effect on the CP of the root soil, which is also shown in the samples containing DR.

Different from the sample containing VR, the TP of Plant B in the devitalised PF–VC sample was slightly larger than that of Plant G. The CP in the VC group increased with Plant G, Plant H and Plant B in turn. Comparing the change trend of porosity before and after root devitalisation, the average value of TP in each sample group decreased from 0.524 to 0.500, and the average value of CP increased from 0.418 to 0.431. The CV values of TP and CP were 0.020 and 0.060, respectively, which were more concentrated than those before devitalisation. The loss of root activity may make the difference in the effect of different plant roots on the porosity of the sample gradually decrease and the effect gradually converge.

3.2.3. SMC

Table 3 shows the SMC of each sample at each growth duration under the influence of different plant roots. The data reveal that with increasing growth duration, the SMC of each plant root sample showed an increasing trend, indicating progressive improvement in water holding capacity. Among them, in the VR-containing samples, the SMC of the samples containing Plant H roots was larger than that of other plants. This phenomenon was reflected in both the PF–VC group and the VC group, potentially related to root density. At 90 d, the maximum SMC of the samples containing Plant H roots exceeded 0.50, indicating that the total water weight in this sample had exceeded the dry weight. At this stage, it had the largest water holding capacity, followed by Plant B, and finally G. In addition, in the VC group, the SMC of each corresponding sample decreased compared with that in the PF–VC group. This indicated that the incorporation of PF had a positive effect on the improvement of SMC in root-containing samples.

Table 3
The change trend of SMC of each sample under different plant roots.

Growth duration (d)	Sample grouping					
	PF–VC group			VC group		
	G	H	B	G	H	B
30 d-(VR)	0.430	0.466	0.483	0.367	0.442	0.420
60 d-(VR)	0.434	0.484	0.487	0.420	0.482	0.456
90 d-(VR)	0.481	0.512	0.491	0.427	0.494	0.464
30 d-(DR)	0.382	0.440	0.394	0.376	0.383	0.403
60 d-(DR)	0.391	0.445	0.434	0.416	0.390	0.403
90 d-(DR)	0.445	0.460	0.465	0.419	0.429	0.405

Note: VR - Samples containing viable roots (VR). DR - Samples containing devitalised roots (DR).

In the DR-containing samples, the SMC of each corresponding sample showed a downward trend compared with that in the VR-containing samples, and showed a certain correlation with the change patterns of BD and TP indicators. In the PF–VC group, the SMC of the Plant H-containing root samples was larger in the early and middle stages, the SMC of the Plant B-containing root samples was larger in the later stage, and the SMC of the G-containing root samples was the smallest in each period. In the VC group, the SMC of the Plant B-containing root samples was the largest in the early stage, and it showed a very stable growth rate with increasing growth duration. The SMC of the Plant H-containing root samples was the largest in the later stage. On the whole, this is related to the differences in root quantity, root distribution and root anti-erodibility of each plant mentioned in the previous analysis.

3.2.4. SPC

As shown in Fig. 10, in the VR-containing sample group, the SPC of samples with different plant roots in the PF–VC group exhibited distinct trends over the growth duration. Among them, Plant G and Plant H showed an upward trend, while Plant B showed a downward trend, but the overall variation range was smaller than that of the VC group. In the VC group, the SPC of each sample decreased rapidly in the early stage and then more slowly with prolonged growth duration. The SPC at 30 d was significantly higher than that of the PF–VC group, whereas at 90 d, it was significantly lower with the difference between these two durations approached an order of magnitude. This indicates that PF showed obvious anti-seepage effects in the early stage. Moreover, the roots of the three plant species significantly reduced SPC throughout the growth duration. The differences among root-containing samples in the PF–VC group can be attributed to the superior structural stability of Plant B's roots. Combined with the early-stage anti-seepage effect of PF, these samples thus exhibited a gradual downward trend as the growth duration increased.

The SPC of samples with H roots was significantly higher than that of the other two plants. This occurs because Plant H's root morphology is primarily axial root type, and the vertical root system strengthens the preferential flow effect during infiltration. Conversely, fibrous root systems extend along slopes to encase soil particles, limiting crack development and thereby reducing permeability. The more developed the fibrous roots, the more pronounced this effect becomes. This may also explain the SPC increase in Plant H-root samples from the PF–VC group, though the change magnitude was small. The small change

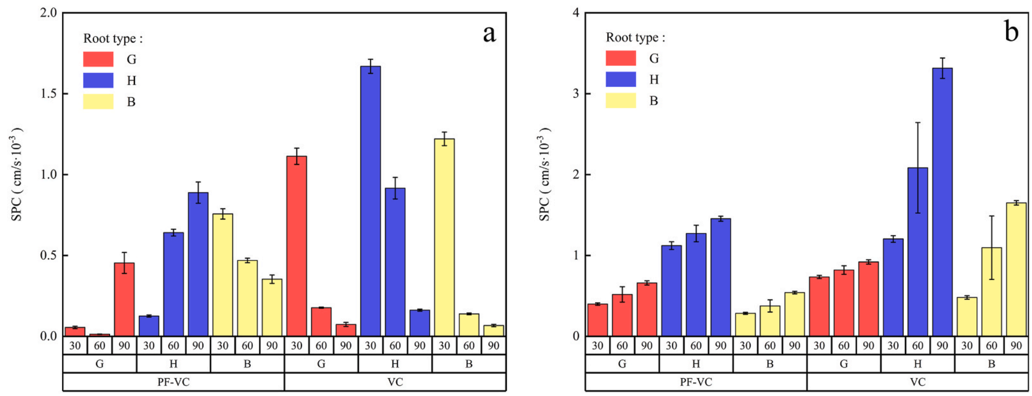

Fig. 10. The change trend of SPC of each sample under different plant roots. a: Viable root-treated sample groups. b: Devitalised root-treated sample groups.

magnitude is consistent with PF becaming black, thin, and brittle in later growth stages, indicating corrosion and decomposition. The additional voids would form water-transport pathways, thereby promoting infiltration in areas where roots were not intertwined.

In the DR-containing sample group, SPC showed a slight increasing trend with growth duration. All VC group samples had higher SPC than PF–VC counterparts, with Plant H-root samples showing the largest increase. The further confirms PF's anti-seepage effect and the enhanced permeability associated with devitalised roots. Among the three plants, Plant B-root samples had the smallest SPC, which correlated with Plant B's slow decomposition and minimal disturbance to soil pore structure. The study by Bormann H and Klaassen K also demonstrated that soil's saturated hydraulic conductivity decreases during rapid root growth and increases with root decay (Bormann and Klaassen, 2008).

3.3. Soil-water characteristic curve (SWCC)

The SWCC constitutes a fundamental constitutive relation in unsaturated soil mechanics. It describes the relationship between matrix suction and water content in soils, effectively reflecting soil water holding capacity. SWCCs for unsaturated soils are established using the logarithm of matrix suction as the abscissa. From left to right, the curve evolves through three characteristic zones: the boundary effect zone, transition zone, and unsaturated residual zone, each exhibiting distinct curve morphology.

3.3.1. VR sample group

Fig. 11 shows the SWCC of each sample containing VR at different growth durations. As the growth duration increases, the plant roots gradually extend downward, resulting in differences in SWCC between periods. At 30-day growth duration, plant roots were mainly distributed at 0–2 cm depth, and therefore the SWCC of each sample in the 0–2 cm section was analysed (Fig. 11a). At 60-day growth duration, roots were mainly distributed at 0–4 cm depth, so the SWCC of each sample in the 0–4 cm section was analysed (Fig. 11b and c). At 90-day growth duration, roots were mainly distributed at 0–6 cm depth, and accordingly the SWCC of each sample in the 0–6 cm section was analysed (Fig. 11d–f).

The figure shows that matrix suction of root-containing samples decreases with increasing volumetric moisture content. This phenomenon persists throughout all growth durations, exhibiting a typical three-stage S-shaped distribution: boundary effect zone (0–10^2 kPa), transition zone (10^2–10^4 kPa), and unsaturated residual zone (>10^4 kPa). All curves converge at the inflection point, with variations between curves primarily showing vertical stretching/compression around this central point. The experimentally determined relationship between matrix suction and volumetric moisture content was fitted using the VG model. Fitting parameters and goodness of fit for the 30-, 60- and 90-day growth durations are presented in Tables 4–6. The goodness of fit (R^2) exceeded 0.9 for all samples, with a mean > 0.96, indicating good model fit. The occurrence of zero values in residual volumetric moisture content for a few instances may be attributed to the inherently low water holding capacity of the sandy loam soil (Ng and Peprah-Manu, 2023).

α, defined as the reciprocal of the air-entry value, relates to soil pore size. The magnitude of α indicates soil water holding capacity: the smaller the α value, the stronger the water holding capacity in the low-suction section (Ip et al., 2019). After 30 days' growth, α values for samples Plant G, Plant H and Plant B in the PF–VC group were 2.04, 1.75 and 1.09 times those of the CK (Z) group, respectively. In the VC group, α values for Plant G, Plant H and Plant B were 23 %, 36 % and 33 % of those in the CK (W) group, respectively; Plant H exhibited the highest α value (0.0045). PF incorporation decreased the α value. The α value also decreased under the sole action of plant roots. When roots and fibre acted in combination, the α value increased. Root growth thus counteracted the fibre-induced enhancement of water holding capacity. Nevertheless, compared with CK (W), α still showed a decreasing trend, indicating sustained enhancement of water holding capacity in the low-suction section.

Defined as the slope of the transition zone, n relates to pore size distribution. The smaller the n value, the wider the pore size distribution in the sample and the weaker the water holding capacity in the high-suction section (Liu et al., 2025a). Overall, the CK (W) group exhibited the lowest n value. This indicates that both individual and combined applications of PF and plant roots enhanced the water holding capacity of the high-suction section compared to samples without fibre or roots. In the PF–VC group, n values were smaller than those of the CK (Z) group and increased progressively in the sequence of samples Plant G, Plant H, and Plant B. This occurred because media particles in PF-containing samples closely bonded closely, resulting in abundant small pores. Roots extended into pores and filled them, thereby reconstructing surrounding soil particle structures and increasing pore anisotropy (Helliwell et al., 2017). Among these, Plant G with rapid growth and the most abundant fibrous roots showed the most significant disturbance to soil particle structures and pore-filling effect, making pore size distribution more dispersed. While fibre incorporation enhanced water holding capacity in high-suction sections, root synergy weakened this effect. In contrast, within the VC group, n values surpassed those of the CK (W) group and decreased progressively in the order of samples Plant G, Plant H, and Plant B. Under the sole influence of plant roots, the water holding capacity of each sample in the high-suction section was enhanced. This phenomenon can be attributed to the relatively compacted soil structure in the VC group, where the distribution of rice husk (RH) and organic matter created abundant pores. These pores functioned as localised water sources by absorbing and retaining moisture, regulating the water content in surrounding soil particles, and guiding inward root growth. As roots penetrated and interpenetrated the soil, they filled existing pores, thereby increasing the number of small pores (Dimattia et al., 2025). Simultaneously, both the air-entry value and slope n increased due to root water absorption. Notably, comparative

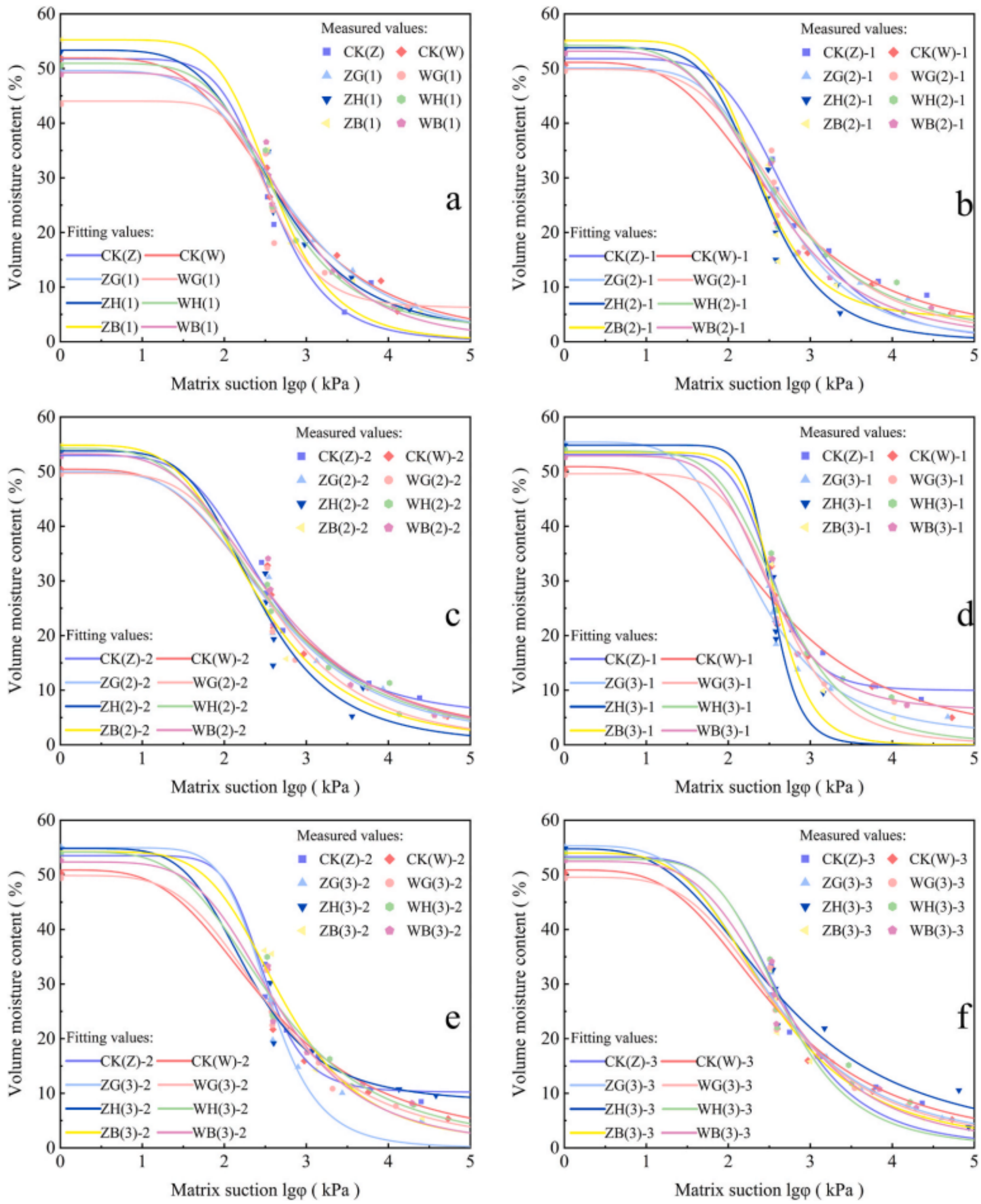

Fig. 11. SWCC of samples containing viable roots (VR) from various plant species at different growth durations. CK- (Z) was a PF–VC control group without plant growth. CK- (W) was the VC control group without plant growth. Z and W represent PF–VC group and VC group, respectively. Cynodon dactylon (G), Lolium perenne (H), Paspalum notatum (B). The numbers 1, 2 and 3 in the brackets represent the growth duration of 30 d, 60 d and 90 d, respectively. -1, -2, -3 represent the depth of 0–2 cm, 2–4 cm, 4–6 cm respectively.

analysis between the PF–VC and VC groups revealed that sample ZB (1), unlike samples Plant G and Plant H, exhibited a smaller α value but a larger n value than WB (1) This indicates that the synergistic interaction between fibres and Plant B roots enhanced water holding capacity, whereas combinations of fibres with Plant G or Plant H roots yielded adverse effects.

As the growth duration advanced to 60 days, the roots further penetrated into the soil layer. In the 0–2 cm soil layer, the α value of each sample in the PF–VC group was lower than that at 30 d. The overall value was smaller than that in the CK (Z)-1 group, while the n value was relatively increased, reaching a level comparable to that in the CK (Z)-1 group. At this time, the addition of fibre and plant growth could further improve the water holding capacity of the sample. The α value of each sample in the VC group increased compared with that at 30 d, rising to 33 %, 58 %, and 36 % of CK (W)-1 in the order of Plant G, Plant H, and Plant B, exceeding the PF–VC group. This may be due to the fact that the vertical depth of plant roots broke through the height of the test soil sample, where increased root presence enhanced pore connectivity. This enhanced connectivity reduced water retention suction, consequently decreasing the air-entry value. The n value of Plant G and Plant H samples decreased compared with that at 30 d, while the n value of Plant B increased relatively. However, they were lower than those of the PF–VC group, indicating that the combination of fibre and plant root system had a stronger effect on increasing the water holding capacity of the sample than the individual effects of plant root system and fibre alone. In both groups, the α value of Plant H was the largest. Its thicker root diameter and vertical distribution facilitated more interconnected pores, easing drainage and reducing the air-entry value. The n value of Plant B was the largest in each group, and its thick, obliquely growing roots intertwined within the sample, promoting a more uniform pore

Table 4
Hydraulic characteristic parameters of each sample when the growth duration is 30 d.

Sample group	θ_s	θ_r	α/kPa^{-1}	n	R^2
CK(Z)	0.52	0.081	0.0034	4.48	0.93
CK(W)	0.52	0	0.0124	1.37	0.98
ZG (1)	0.49	0.037	0.0069	1.53	0.97
ZH (1)	0.53	0.056	0.0059	1.77	0.97
ZB (1)	0.55	0.082	0.0037	2.88	0.98
WG (1)	0.43	0.094	0.0028	8.29	0.97
WH (1)	0.51	0.066	0.0045	2.03	0.98
WB (1)	0.49	0.063	0.0041	2.00	0.97

Note: CK- (Z) was a PF–VC control group without plant growth. CK- (W) was the VC control group without plant growth. Z and W represent PF–VC group and VC group, respectively. *Cynodon dactylon* (G), *Lolium perenne* (H), *Paspalum notatum* (B). The numbers 1, 2 and 3 in the brackets represent the growth duration of 30 d, 60 d and 90 d, respectively.

Table 5
Hydraulic characteristic parameters of each sample when the growth duration is 60 d.

Sample group	θ_s	θ_r	α/kPa^{-1}	n	R_2
CK(Z)-1	0.53	0.073	0.0037	4.66	0.98
CK(W)-1	0.51	0	0.0124	1.42	0.98
ZG (2)-1	0.50	0.099	0.0033	4.12	0.98
ZH (2)-1	0.54	0.078	0.0036	5.50	0.97
ZB (2)-1	0.55	0.077	0.0033	6.59	0.99
WG (2)-1	0.50	0.074	0.0041	2.18	0.96
WH (2)-1	0.54	0.07	0.0072	1.78	0.96
WB (2)-1	0.53	0.073	0.0045	2.37	0.98
CK(Z)-2	0.53	0.063	0.0044	4.96	0.98
CK(W)-2	0.50	0	0.011	1.35	0.95
ZG (2)-2	0.50	0.061	0.0088	1.63	0.97
ZH (2)-2	0.54	0.079	0.0033	6.36	0.98
ZB (2)-2	0.55	0.079	0.0045	2.91	0.99
WG (2)-2	0.50	0.081	0.0037	2.94	0.96
WH (2)-2	0.54	0.048	0.012	1.53	0.99
WB (2)-2	0.53	0.034	0.0087	1.54	0.96

Note: CK- (Z) was a PF–VC control group without plant growth. CK- (W) was the VC control group without plant growth. Z and W represent PF–VC group and VC group, respectively. *Cynodon dactylon* (G), *Lolium perenne* (H), *Paspalum notatum* (B). The numbers 1, 2 and 3 in the brackets represent the growth duration of 30 d, 60 d and 90 d, respectively. −1, −2, −3 represent the depth of 0–2 cm, 2–4 cm, 4–6 cm respectively.

size distribution.

When soil depth reached 2–4 cm, the α value in the PF–VC group showed an upward trend versus the upper layer, with Plant G exhibiting the largest value. This trend mirrored PF–VC group's variation versus CK at 30-day growth stage, confirming that root penetration into specific soil layers and its involvement in water migration induces consistent alterations in air-entry value. Regarding n values, Plant G and Plant B were significantly smaller than those in the upper layer, whereas Plant H increased further and was significantly larger than the other two. As soil depth increased, fibrous root distribution decreased substantially, and deeper roots developed thicker diameters. Plant H, characterized by an axial root system, exhibited the thickest roots. Studies suggest that thicker roots correlate with more macropores in the rhizosphere, significantly raising n values (Du et al., 2025). The VC group showed species-specific patterns. Plant G's α value trend was consistent with that at 30-day, whereas Plant H and Plant B increased significantly. Plant H's maximum α value was 0.0121, approaching the CK group (0.0124). In the 2–4 cm layer, Plant H's low-suction water holding capacity weakened significantly, however, fibre addition reduced its α value, indicating that fibre-Plant H root synergy enhanced low-suction capacity. n value trends mirrored 30-day data: all VC samples exceeded the CK group's value (1.35), with Plant G exhibiting a higher n value than the upper soil layer, while Plant H and Plant B were lower. Differential growth distributions and root traits (comparing Plant G with Plant H and Plant B) governed pore structure and water holding capacity variations.

Table 6
Hydraulic characteristic parameters of each sample when the growth duration is 90 d.

Sample group	θ_s	θ_r	α/kPa^{-1}	n	R^2
CK(Z)-1	0.53	0.067	0.0039	4.89	0.97
CK(W)-1	0.50	0	0.011	1.45	0.93
ZG (3)-1	0.55	0.067	0.0031	2.55	0.98
ZH (3)-1	0.55	0.094	0.0029	15.27	0.99
ZB (3)-1	0.54	0.074	0.0030	6.10	0.99
WG (3)-1	0.49	0.089	0.0035	3.06	0.98
WH (3)-1	0.54	0.11	0.0039	2.87	0.98
WB (3)-1	0.53	0.092	0.0031	3.04	0.97
CK(Z)-2	0.54	0.054	0.0047	5.24	0.97
CK(W)-2	0.50	0	0.0100	1.38	0.92
ZG (3)-2	0.55	0.071	0.0044	2.57	0.97
ZH (3)-2	0.55	0.088	0.014	1.68	0.96
ZB (3)-2	0.54	0.059	0.0044	2.04	0.97
WG (3)-2	0.49	0.071	0.0057	1.90	0.97
WH (3)-2	0.54	0.069	0.0082	1.67	0.96
WB (3)-2	0.53	0.086	0.0061	1.91	0.97
CK(Z)-3	0.54	0.053	0.0047	5.65	0.92
CK(W)-3	0.50	0	0.010	1.35	0.92
ZG (3)-3	0.55	0.027	0.013	1.46	0.98
ZH (3)-3	0.55	0	0.015	1.25	0.96
ZB (3)-3	0.54	0.061	0.0064	1.86	0.97
WG (3)-3	0.49	0.064	0.0077	1.66	0.96
WH (3)-3	0.54	0.12	0.013	6.36	0.98
WB (3)-3	0.53	0.074	0.0066	1.99	0.97

Note: CK- (Z) was a PF–VC control group without plant growth. CK- (W) was the VC control group without plant growth. Z and W represent PF–VC group and VC group, respectively. *Cynodon dactylon* (G), *Lolium perenne* (H), *Paspalum notatum* (B). The numbers 1, 2 and 3 in the brackets represent the growth duration of 30 d, 60 d and 90 d, respectively. −1, −2, −3 represent the depth of 0–2 cm, 2–4 cm, 4–6 cm respectively.

After 90 days of growth, plant root depth deepened to three sample heights, prompting stratified sampling into three layers. Compared with the 60 d growth duration, the α value of the upper layer in both the PF–VC group and the VC group decreased further, and the water holding capacity of the low-suction section in each sample's soil layer was enhanced. In terms of the n value, samples with different plant roots in the PF–VC group showed distinct trends. The Plant H root system and fibre exhibited a synergistic enhancement effect in the upper soil layer: the n value of the ZH (3)-1 group was the largest (15.27). In contrast, the VC group's n values showed an overall upward trend, refining a more uniform pore size distribution among the samples at this stage.

Compared with the 60 d growth duration, the α values of the Plant G and Plant B in the PF–VC group's middle soil layer decreased, with the Plant G's α value decreasing by 50 %, while the Plant H's α value increased to exceed that of the CK group. The Plant G's n value increased, whereas the n values of the Plant H and Plant B decreased; the H sample's value decreased by 78.3 % compared with that at 60 d. This may be attributed to the further development and extension of the fibrous root system in this layer. Root parameters distribution trends indicated that the root volume of the Plant H was substantially larger than that of other species, yet its fibrous roots at this depth were sparse. Coarse roots penetrate the middle soil layer and produce extrusion disturbance, leading to pore size redistribution, which is related to the significant reduction in the air-entry value and n value. In the VC group, each plant's root system continued to develop into the deep layer. Differences between the Plant G root system and other plants' root distribution depth led to an increase in the Plant G's α value and a decrease in those of the Plant H and Plant B samples, with corresponding decreases and increases in their n values.

In the lowest soil layer, the α values of both the PF–VC and VC groups increased within the depth range; owing to the limited root abundance and larger root diameters, these values were the largest, with the PF–VC group showing a greater increase with depth than the VC group. The n value of each sample in the PF–VC group decreased with increasing soil depth, and the Plant B exhibited the best water holding capacity in this section. In the VC group, the Plant G's n value decreased with increasing soil depth, while those of the Plant H and Plant B increased. The Plant B's minimum α value corresponds to strong water holding capacity in the low-suction section. The Plant H's n value (6.36) is significantly larger than that of other plants, indicating the strongest water holding capacity in the high-suction section of this soil layer.

3.3.2. DR sample group

The SWCC of samples treated with plant root devitalisation at each growth duration, obtained via fitting, is shown in Fig. 12. Among them, the SWCC of 0–2 cm soil layer for each sample at 30 d growth duration is presented in Fig. 12a. The SWCC of 0–4 cm soil layer for each sample at 60 d growth duration is shown in Fig. 12b and c. The SWCC of 0–6 cm soil layer for each sample at 90 d growth duration is illustrated in Fig. 12d–f. At 30 d growth duration, one or two groups of discrete points were observed in the fitting curve of each sample group, whereas the curve fitting was good at other growth stages.

It can be seen from the three stages of the curve that in the boundary effect zone (0–10^2 kPa), the SWCC of each sample after plant root devitalisation is located below that of the control group. The slope of the transition zone (10^2 kPa–10^4 kPa) is slower than in the control group and the unsaturated residual zone (>10^4 kPa) is located above that of the control group. Among them, the unsaturated residual zone of the PF–VC group was at the bottom. The fitted parameter results and goodness of fit are shown in Tables 7–9. The goodness of fit was 71–89 %

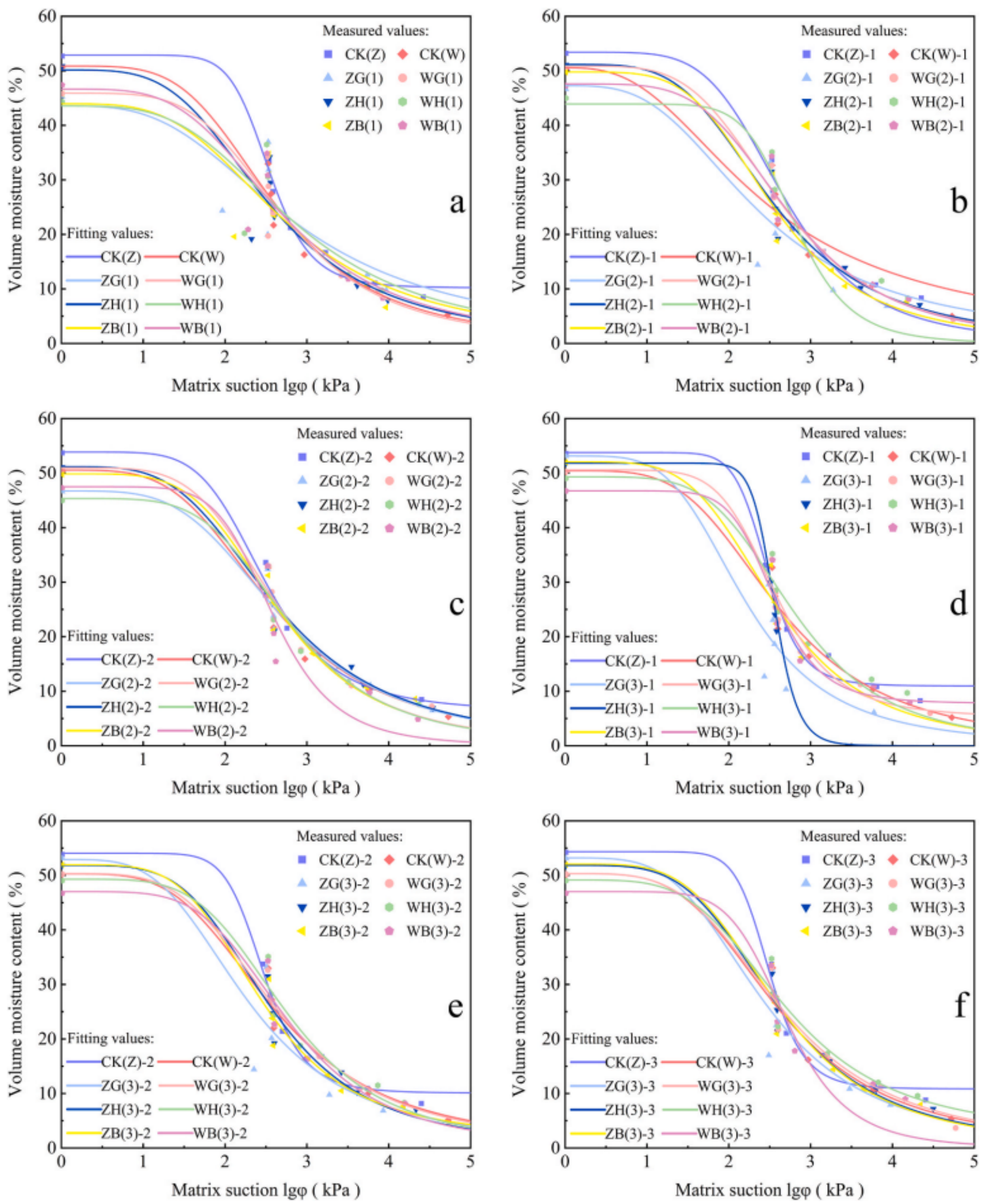

Fig. 12. SWCC of samples containing devitalised roots (DR) from various plant species at different growth durations. CK- (Z) was a PF–VC control group without plant growth. CK- (W) was the VC control group without plant growth. Z and W represent PF–VC group and VC group, respectively. Cynodon dactylon (G), Lolium perenne (H), Paspalum notatum (B). The numbers 1, 2 and 3 in the brackets represent the growth duration of 30 d, 60 d and 90 d, respectively. -1, -2, -3 represent the depth of 0–2 cm, 2–4 cm, 4–6 cm respectively.

Table 7
Hydraulic characteristic parameters of devitalised treatment samples when the growth duration is 30 d.

Sample group	θ_s	θ_r	α/kPa^{-1}	n	R^2
CK(Z)	0.53	0.073	0.0037	4.66	0.98
CK(W)	0.51	0	0.012	1.42	0.98
ZG (1)	0.45	0	0.055	1.20	0.72
ZH (1)	0.51	0	0.032	1.29	0.84
ZB (1)	0.46	0	0.035	1.25	0.74
WG (1)	0.46	0	0.013	1.59	0.89
WH (1)	0.44	0	0.019	1.27	0.75
WB (1)	0.47	0	0.020	1.30	0.83

Note: CK- (Z) was a PF–VC control group without plant growth. CK- (W) was the VC control group without plant growth. Z and W represent PF–VC group and VC group, respectively. *Cynodon dactylon* (G), *Lolium perenne* (H), *Paspalum notatum* (B). The numbers 1, 2 and 3 in the brackets represent the growth duration of 30 d, 60 d and 90 d, respectively.

Table 8
Hydraulic characteristic parameters of devitalised treatment samples when the growth duration is 60 d.

Sample group	θ_s	θ_r	α/kPa^{-1}	n	R_2
CK(Z)-1	0.53	0.067	0.0039	4.89	0.97
CK(W)-1	0.5	0	0.011	1.45	0.93
ZG (2)-1	0.47	0.044	0.0059	1.61	0.95
ZH (2)-1	0.51	0.058	0.0075	1.65	0.96
ZB (2)-1	0.5	0.099	0.0044	2.68	0.98
WG (2)-1	0.51	0.076	0.0050	2.01	0.98
WH (2)-1	0.45	0.098	0.0029	12.92	0.96
WB (2)-1	0.47	0.087	0.0030	7.23	0.92
CK(Z)-2	0.54	0.054	0.0047	5.24	0.97
CK(W)-2	0.5	0	0.010	1.38	0.92
ZG (2)-2	0.47	0.046	0.0098	1.49	0.96
ZH (2)-2	0.51	0.045	0.013	1.45	0.96
ZB (2)-2	0.5	0.082	0.0070	1.79	0.97
WG (2)-2	0.51	0.078	0.0048	2.06	0.98
WH (2)-2	0.45	0.084	0.0040	2.15	0.97
WB (2)-2	0.47	0.073	0.0029	7.43	0.98

Note: CK- (Z) was a PF–VC control group without plant growth. CK- (W) was the VC control group without plant growth. Z and W represent PF–VC group and VC group, respectively. *Cynodon dactylon* (G), *Lolium perenne* (H), *Paspalum notatum* (B). The numbers 1, 2 and 3 in the brackets represent the growth duration of 30 d, 60 d and 90 d, respectively. −1, −2, −3 represent the depth of 0–2 cm, 2–4 cm, 4–6 cm respectively.

at 30 d, 91–98 % at 60 d, and 82–99 % at 90 d, indicating better fitting in the later stages while maintaining acceptable accuracy throughout. The phenomenon of zero residual volumetric moisture content in samples after root devitalisation treatment is more common than in viable root treatment samples. This is because root pore channels form preferential paths, but these paths diminish with increasing depth (Leung et al., 2018).

When comparing devitalised root-treated with viable root-treated samples at the 30-day growth duration, it was found that except for the CK (Z) and CK (W) control groups, the α values of all samples increased significantly and were greater than those in the CK group. This indicates that the pore size distribution of samples subjected to root devitalisation treatment is wider, and water holding capacity is significantly weakened. Among them, the α value of the PF–VC group was significantly larger than that of the VC group, and the n value was slightly smaller than that of the VC group. This was because PF incorporation altered the pore structure of VC, thereby influencing plant growth and subsequent root parameters such as root diameter and root volume. The PF–VC group provided more space for the colonization of plant roots. During the decomposition of devitalised roots, the decomposition rate of fine roots was faster than that of coarse roots, and their number decreased more (Tunc and von Wiren, 2025). Therefore, the

Table 9
Hydraulic characteristic parameters of devitalised treatment samples when the growth duration is 90 d.

Sample group	θ_s	θ_r	α/kPa^{-1}	n	R^2
CK(Z)-1	0.54	0.063	0.0041	5.19	0.98
CK(W)-1	0.50	0	0.010	1.47	0.94
ZG (3)-1	0.53	0.04	0.013	1.78	0.86
ZH (3)-1	0.52	0.16	0.0033	8.28	0.99
ZB (3)-1	0.52	0.020	0.0046	2.02	0.98
WG (3)-1	0.50	0.083	0.0037	2.55	0.97
WH (3)-1	0.49	0.11	0.0040	2.30	0.97
WB (3)-1	0.47	0.10	0.0037	3.97	0.96
CK(Z)-2	0.54	0.049	0.0050	5.38	0.98
CK(W)-2	0.50	0	0.010	1.43	0.95
ZG (3)-2	0.53	0	0.0044	1.33	0.83
ZH (3)-2	0.52	0.067	0.015	1.61	0.96
ZB (3)-2	0.52	0.082	0.0064	1.94	0.96
WG (3)-2	0.50	0.020	0.0096	1.45	0.97
WH (3)-2	0.49	0.097	0.0090	2.30	0.97
WB (3)-2	0.47	0.093	0.0089	2.32	0.96
CK(Z)-3	0.54	0.045	0.0054	5.73	0.97
CK(W)-3	0.50	0	0.0098	1.37	0.94
ZG (3)-3	0.53	0	0.014	1.35	0.9
ZH (3)-3	0.52	0.057	0.018	1.54	0.97
ZB (3)-3	0.52	0.081	0.0066	1.85	0.96
WG (3)-3	0.50	0	0.014	1.36	0.97
WH (3)-3	0.49	0.096	0.0060	1.81	0.95
WB (3)-3	0.47	0.11	0.0033	3.40	0.98

Note: CK- (Z) was a PF–VC control group without plant growth. CK- (W) was the VC control group without plant growth. Z and W represent PF–VC group and VC group, respectively. *Cynodon dactylon* (G), *Lolium perenne* (H), *Paspalum notatum* (B). The numbers 1, 2 and 3 in the brackets represent the growth duration of 30 d, 60 d and 90 d, respectively. −1, −2, −3 represent the depth of 0–2 cm, 2–4 cm, 4–6 cm respectively.

root pores formed by root decomposition in the PF–VC group were more numerous and larger. At the same time, PF, as an organic substance, although its natural degradation rate was slow, also caused the redistribution of pore structure. The decrease in n value also indicated that the breadth of pore distribution increased. Differences between plants are due to differences in root parameters and texture. For example, the n value of WG (1) in the VC group is larger than that of the other two groups. This is because more fibrous roots of the Plant G root system decompose to produce or expand pores, and the combination of pores leads to a more uniform pore size distribution.

In the 0–2 cm soil layer, compared with devitalised root-treated samples at 30 d growth duration, the α values of the PF-VC and VC groups decreased at 60 d, while the n values increased; indicating partial recovery of water holding capacity in devitalised root-treated samples. The α and n values of devitalised root-treated samples at 60 d growth duration showed different trends compared with viable root-treated samples in the same period. The α value of the PF-VC group increased and the n value decreased. The increase in α value was much smaller than that at 30 d, while the decrease in n value was significantly greater. This indicates that the deterioration of water holding capacity of samples was suppressed at 60 d. A large number of root pores form after the decomposition of devitalised roots, but organic substances released during decomposition (such as polysaccharides and proteins) can bind to soil mineral particles to promote the formation of soil aggregates. Overall, although water holding capacity remained lower than in viable root-treated samples, pore stability was partly maintained (Zhai et al., 2025). The α value of Plant G root in the VC group increased and the n value decreased, with a small range of variation. The other two plant root types showed opposite trends in α and n values, with a larger amplitude of change; notably, the n value of Plant H root increased significantly to 12.92. The rapid decomposition of more fibrous roots in Plant G reduced the sample water holding capacity. The VC group had a compact, hard structure with less root distribution. The thicker roots of

Plant H and Plant B were thicker likely contributed to the sharp *n* values increase in these samples. With increasing soil depth, the *α* value of the PF-VC group increased and the *n* value decreased. Changes in the VC group values were not obvious, while the *n* value of Plant H root decreased sharply. Studies indicate that higher humidity in deeper layers enhances microbial biomass and enzyme activity, accelerating the decomposition of roots and PF compared to surface layers. This accelerates the alteration of pore characteristics in deeper soils (Geng et al., 2025).

At 90 days of growth duration, when comparing devitalised root-treated samples with viable root-treated samples, the overall trends of their *α* and *n* values were similar to the results of the aforementioned comparative analysis. However, in the VC group, the *n* value of the WB (3)-1 group in the 0–2 cm soil layer increased. In the 2–4 cm soil layer, the *n* values of the WH(3)-2 and WB(3)-2 groups increased and in the 4–6 cm soil layer, the *α* values of the WH(3)-3 and WB(3)-3 groups

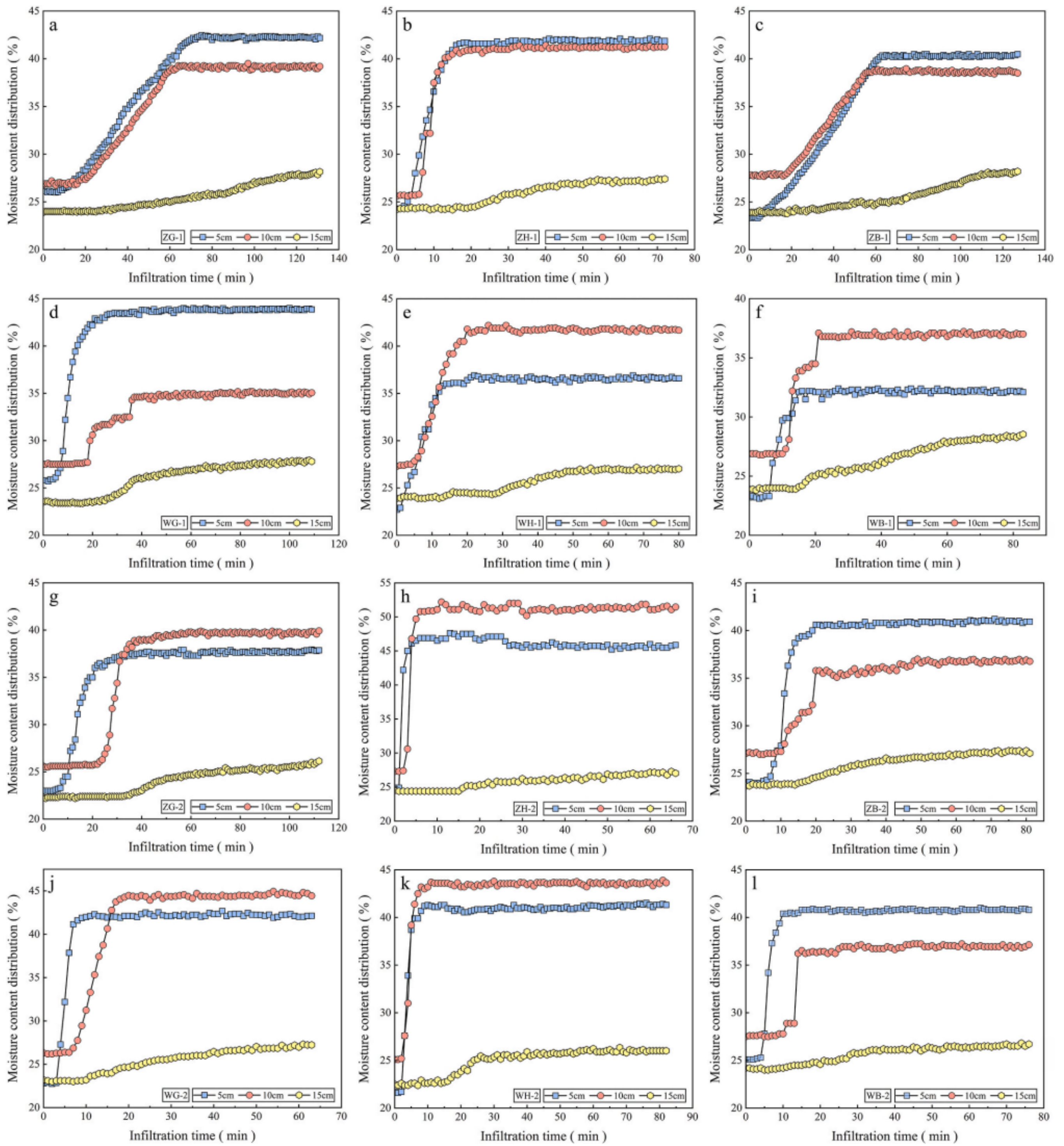

Fig. 13. Variation of moisture content at different depths in viable root-treated sample columns. a–f: 30 d growth duration. g–l: 60 d growth duration. m–r: 90 d growth duration. Z and W represent PF–VC group and VC group, respectively. Cynodon dactylon (G), Lolium perenne (H), Paspalum notatum (B). The numbers 1, 2 and 3 represent the growth duration of 30 d, 60 d and 90 d, respectively.

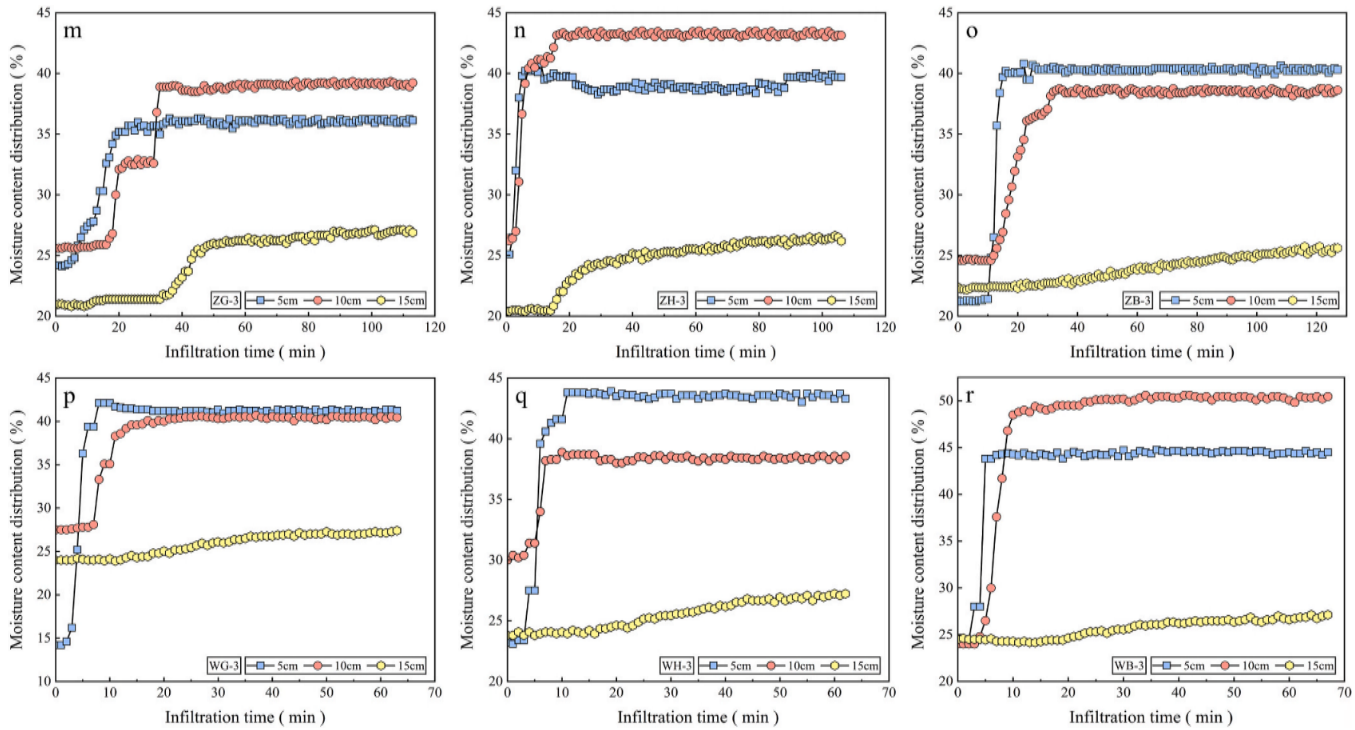

Fig. 13. (*continued*).

decreased while the *n* value of the WB(3)-3 group increased. This is attributed to the decomposition resistance and large biomass of Plant B roots, which significantly enhance structural heterogeneity across all samples. Observations of PF during sampling revealed that the decay cycle of PF following self-incorporation was longer than that of devitalised roots, and the n value of Plant B root in the VC group increased rather than decreased.

3.4. Soil column infiltration characteristics

3.4.1. VR sample group

In the process of the soil column infiltration test, when the wetting front reaches a certain depth, the moisture content at this position changes sharply. The time from water infiltration to the increase in soil moisture content is the time at which the wetting front reaches this depth. The change in the moisture content of the soil column reflects, to a certain extent, a comprehensive index of infiltration. As shown in Fig. 13, the variation trend of moisture content at different depths of viable root-treated sample columns with infiltration time under one-dimensional infiltration conditions is presented. Herein, the moisture content is the mass moisture content, and the infiltration time continues until water leakage occurs at the bottom of the sample column.

It can be seen from the figure that the moisture content variation trend is similar across sample groups, but differs at different depths—this is particularly evident at the stratified positions of the soil column. The moisture content of the VC layer in the upper part of the soil column shows a trend of brief stabilization, followed by a rapid rise and subsequent stabilization over time. Notably, the response in the deeper layer (10 cm) lags behind that in the shallow layer (5 cm). At the beginning of constant head infiltration, soil saturation is low, matrix suction is high, seepage velocity is faster, and moisture content increases rapidly. As seepage proceeds, soil saturation rises, matrix suction decreases, and the rate of increase in moisture content gradually slows down, eventually stabilising. The lower ES layer has higher matrix suction; water is thus slowly drawn from the upper VC layer via water potential, maintaining a trend of gradual rise. From the perspective of initial moisture content, the initial moisture content in the VC layer increases with depth. This is because when the wetting front reaches 10 cm, the upper soil is fully wetted, forming a stable high hydraulic conductivity zone. The strong water conductivity of the upper water pressure and wetted zone imposes a stronger hydraulic driving potential on the deep layer (Ramanathan et al., 2023). Additionally, gas in the deep soil is difficult to expel and becomes compressed to form a gas barrier, meaning the wetting front requires higher air pressure to continue infiltrating. This causes water to accumulate at the front, leading to an increase in initial moisture content (Cui et al., 2025). The change in initial moisture content at 15 cm arises from the further integration of performance differences between different materials.

The onset time of moisture increase, time to peak moisture content, and total infiltration duration for each group of soil column samples are presented in Table 10.

At 30 d, in the PF–VC group and the VC group, the total infiltration duration of the Plant G root was the longest, followed by the Plant B root, with the Plant H root exhibiting the shortest total infiltration duration. At 60 d, the PF–VC group maintained this trend, while in the VC group, the Plant G root had the shortest total infiltration time, the Plant H root had the longest, and the Plant B root was intermediate. At 90 d, in both the PF–VC group and the VC group, the Plant B root showed the longest total infiltration time, followed by the Plant G root, with the Plant H root again having the shortest. Different plant species exert varying effects across different periods. We found that the total infiltration time for most samples in the PF–VC group was longer than that in the VC group. The only exception was the Plant H root, which had a shorter infiltration time at 30 and 60 days. This indicates that the incorporation of PF has a positive effect on delaying water infiltration. With increasing growth duration, differences emerged in the total infiltration time of each sample within the PF–VC group. Specifically, the total infiltration time of the Plant G root decreased, that of the Plant H root increased, and that of the Plant B root decreased initially before increasing, remaining stable overall. In contrast, the total infiltration time of each sample in the VC group showed an overall decreasing trend.

From the perspective of the time nodes of moisture content change at each depth, a similar trend to the above is observed. The change time at 5 cm occurs within 10 min. The time elapsed in the 5–10 cm section is

Table 10
Time nodes of moisture content variation at different depths in viable root-treated sample columns.

Sample group	Onset time of moisture increase (min)			Time to peak moisture content (min)		Total infiltration duration (min)
	5 cm	10 cm	15 cm	5 cm	10 cm	
ZG-1	7	16	27	73	65	131
ZH-1	2	6	15	16	18	72
ZB-1	5	17	16	63	55	127
WG-1	4	18	23	27	37	109
WH-1	1	4	13	13	20	81
WB-1	6	10	15	15	22	83
ZG-2	7	22	35	23	39	113
ZH-2	3	4	15	6	8	66
ZB-2	4	10	14	16	20	81
WG-2	3	6	10	10	19	63
WH-2	2	2	16	10	11	82
WB-2	4	8	16	10	15	76
ZG-3	4	16	33	20	33	113
ZH-3	1	2	14	6	8	106
ZB-3	10	11	26	17	33	127
WG-3	2	7	12	8	14	63
WH-3	1	2	13	11	8	62
WB-3	2	3	15	5	10	67

Note: Z and W represent PF–VC group and VC group, respectively. *Cynodon dactylon* (G), *Lolium perenne* (H), *Paspalum notatum* (B). The numbers 1, 2 and 3 represent the growth duration of 30 d, 60 d and 90 d, respectively.

generally longer than that in the 0–5 cm section and this also confirms the causal analysis regarding the higher initial moisture content in deeper layers. Overall, the moisture content changes of Plant H root showed the fastest response (1–7 min), followed by Plant B root (2–15 min), with Plant G root responding last (2–22 min). Notably, the VC group as a whole exhibited earlier response than the PF–VC group, and these response times advanced progressively with increasing growth duration. This indicates that the water infiltration rates of the three-plant root-treated samples follow the order: Plant H > Plant B > Plant G. Root development over time promotes water infiltration and accelerates the infiltration process in the VC layer, while the incorporation of PF exerts a delaying effect (Pierret and Lacombe, 2018). Throughout the process, the ES layer shows a slight lag and no obvious peak performance.

3.4.2. DR sample group

The variation trend of moisture content at different depths in devitalised root-treated sample columns with infiltration time under one-dimensional infiltration conditions is shown in Fig. 14. From the changes in moisture content at 5 cm and 10 cm depths in the VC layer, the moisture content curve of each sample over time appears steeper after root devitalisation. This accelerated rate of moisture content increase indicates that devitalised root-treated samples are more conducive to water infiltration than viable root-treated samples. The curve slope of the Plant H root is particularly pronounced, likely because the root channels remaining after the decomposition of deep straight roots are more difficult to fill than surface root channels. In the 15 cm ES layer, there is no significant difference in moisture content changes.

The Onset time of moisture increase, Time to peak moisture content, and Total infiltration duration for each group of devitalised root-treated sample columns are presented in Table 11. When compared with the data of viable root-treated samples, it was found that both the Total infiltration duration and the moisture content response time at each depth were generally advanced, with the Onset time of moisture increase showing an early trend with fluctuations. The Time to peak moisture content was also advanced. Among these, the longest Total infiltration duration was 110 min for ZG-1, which was 16 % lower than the 131 min recorded for viable root-treated ZG-1. The shortest was 45 min for WH-3, representing a 27 % reduction from the 62 min of viable root-treated WH-3. The reduction ratio of the PF–VC group was generally lower than that of the VC group and increased with the extension of growth duration. At this point, the anti-seepage function of PF still exerted a significant effect.

3.5. Correlation analysis (PCA)

Pearson correlation analysis was conducted for each indicator, with the results presented in Fig. 15. In viable root-treated samples, all root morphological parameters exhibited an extremely significant negative correlation with volumetric weight and an extremely significant positive correlation with saturated moisture content. Total porosity and shape coefficient n were highly significantly positively correlated with total root length, root surface area, and root volume. However, no significant correlation was observed between these two parameters and root diameter. Volumetric weight showed an extremely significant negative correlation with total porosity and saturated moisture content, as well as a significant negative correlation with shape coefficient n. Residual moisture content was extremely significantly negatively correlated with the air-entry value correlation coefficient α. Capillary porosity had a significant positive correlation with total root length. The saturated permeability coefficient was extremely significantly positively correlated with non-capillary porosity and extremely significantly negatively correlated with capillary porosity, respectively. The air-entry value correlation coefficient α was significantly negatively correlated with shape coefficient n.

The correlations between the respective indices indicate that root growth reshapes the internal structure of the VC. Specifically, increased root morphological parameters promoted higher capillary porosity and total porosity of the VC, which in turn reduced volumetric weight (Zhu et al., 2022). This promotes the formation of more water-storage pores, resulting in increased saturated moisture content and residual moisture content. Concurrently, the enhanced capillary retardation effect influences permeability, causing a decrease in the saturated permeability coefficient. Furthermore, higher moisture content increases root water absorption intensity and directs roots to extend preferentially towards capillary pores. This promotes further optimization of the pore structure, which aligns with the increasing trend of the shape parameter n (Lei et al., 2023). As mentioned earlier, the morphologies of the soil-water characteristic curves basically converge at the inflection points. This indicates that an increase in the n value is accompanied by a decrease in the air-entry value correlation coefficient α, i.e., demonstrating a negative correlation between them. This observation is also confirmed by the correlation analysis.

When compared with the correlation results of indicators in the viable root-treated sample group, the following new inter-indicator correlations are observed in the devitalised root-treated sample group. Specifically, root diameter exhibits a highly significant positive correlation with total porosity and capillary porosity. The saturated permeability coefficient is significantly positively correlated with total root length, root surface area, and residual moisture content. It is also extremely significantly positively correlated with total porosity and extremely significantly negatively correlated with total infiltration duration. Furthermore, residual moisture content shows an extremely significant positive correlation with total root length, root surface area, and shape parameter n, a significant positive correlation with root volume, and a significant negative correlation with total infiltration duration. The air-entry value correlation coefficient α is extremely significantly positively correlated with total infiltration duration, while total infiltration duration is significantly negatively correlated with total porosity. After root growth and subsequent devitalisation, roots decompose and leave behind root channels. Larger root morphological parameters correspond with greater total porosity. Rapid water infiltration along these root channels results in an increase in the VC's

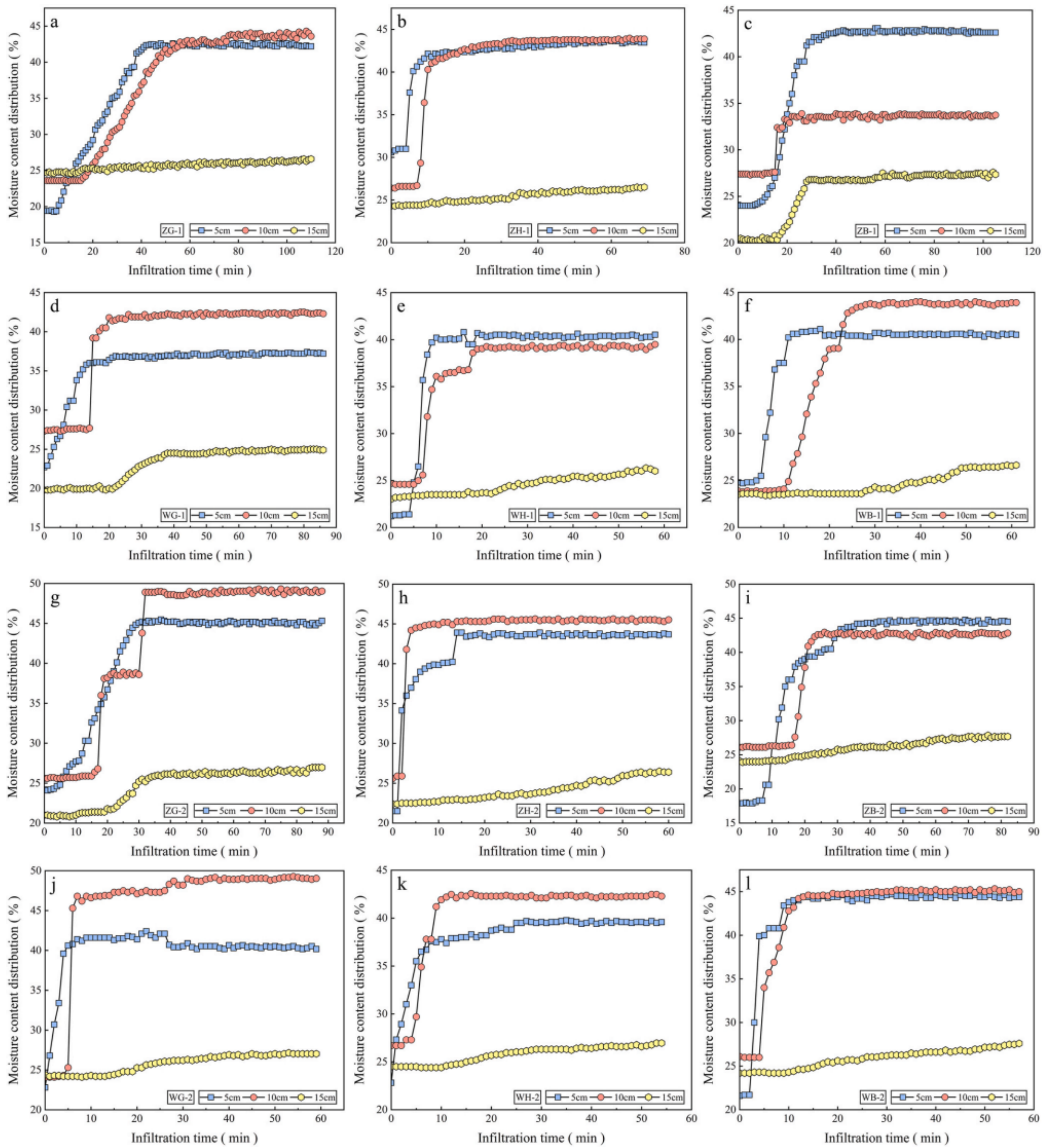

Fig. 14. Variation of moisture content at different depths in devitalised root-treated sample columns. a–f: 30 d growth duration. g–l: 60 d growth duration. m–r: 90 d growth duration. Z and W represent PF–VC group and VC group, respectively. Cynodon dactylon (G), Lolium perenne (H), Paspalum notatum (B). The numbers 1, 2 and 3 represent the growth duration of 30 d, 60 d and 90 d, respectively.

saturated permeability coefficient. Meanwhile, the tiny pores left by fine roots (such as fibrous roots) aid in water retention, thereby increasing the VC's residual moisture content. The high hydraulic conductivity of root channels significantly shortens the total infiltration duration.

4. Discussion

4.1. Fibre incorporation effects

The incorporation of PF improves the pore structure of VC and promotes plant root development. Specifically, the fibers interweave

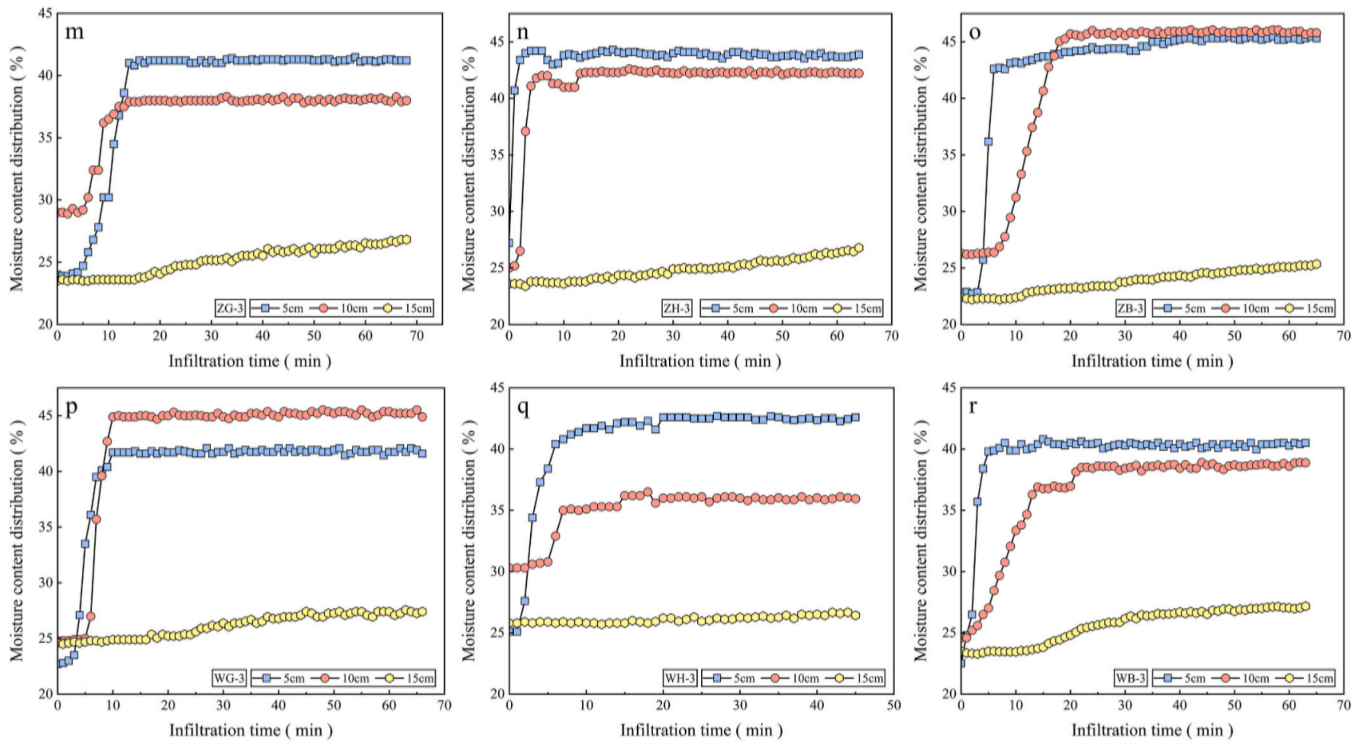

Fig. 14. (*continued*).

Table 11
Time nodes of moisture content variation at different depths in devitalised root-treated sample columns.

Sample group	Onset time of moisture increase (min)			Time to peak moisture content (min)		Total infiltration duration (min)
	5 cm	10 cm	15 cm	5 cm	10 cm	
ZG-1	5	15	26	42	56	110
ZH-1	4	7	10	10	13	69
ZB-1	8	15	16	22	31	105
WG-1	1	14	21	14	20	86
WH-1	3	5	16	10	20	58
WB-1	4	10	27	12	28	61
ZG-2	3	15	19	30	32	88
ZH-2	1	2	16	9	11	60
ZB-2	7	16	24	21	24	82
WG-2	1	4	13	7	9	59
WH-2	1	4	10	8	12	54
WB-2	2	4	9	11	14	57
ZG-3	3	5	15	14	17	68
ZH-3	1	2	14	4	6	64
ZB-3	3	6	11	7	20	65
WG-3	2	5	8	10	12	66
WH-3	1	2	15	6	7	45
WB-3	1	2	12	5	12	63

Note: Z and W represent PF–VC group and VC group, respectively. *Cynodon dactylon* (G), *Lolium perenne* (H), *Paspalum notatum* (B). The numbers 1, 2 and 3 represent the growth duration of 30 d, 60 d and 90 d, respectively.

with, fill, and segment the internal pores of the specimen. VC generally exhibits reduced bulk density, increased porosity, and a more diverse pore size distribution. These observations are further supported by changes in bulk density and porosity observed in control specimens. Additionally, the inclusion of PF reduces soil particle compaction in VC. Collectively, these effects diminish the resistance encountered by plant roots during growth, thereby facilitating root development (Noh et al., 2025). Furthermore, due to PF's pore restructuring effect and strong water absorption capacity, localized water supply points form within the VC, guiding the proliferation of fibrous roots into micro-pores. Furthermore, through ongoing microbial activity within the substrate matrix, PF undergoes natural degradation, thereby providing a sustained supply of organic nutrients for plant growth (Ferreira et al., 2021) (Ogorek et al., 2025). It is worth noting that PF induces the greatest increase in root length for vertical lant H and the largest enhancement in root diameter for coarse plant B. This indicates that the regulatory effects of PF incorporation further amplify the inherent advantages of different plant root systems (see Fig. 16).

In terms of water holding capacity, the incorporation of PF enhances the water storage capacity of VC by improving its pore structure. The abundant hydrophilic functional groups (e.g., carboxyl and hydroxyl groups) on the surface of PF also help immobilize water and form localized water supply points (Célino et al., 2014). Changes in porosity indicators and SWCC parameters both demonstrate an increase in capillary water-retaining pores within VC after PF incorporation, thereby improving its water holding efficiency (Yang et al., 2023). Correspondingly, PF restricts the development of large pores in VC and guides plant roots to grow and fill various pores, a phenomenon that further enhances the water holding capacity of VC. Regarding infiltration performance, although an initial increase in total porosity of VC is observed upon PF incorporation, the fibers actually hinder pore connectivity or segment already connected pores. This alteration complicates the water infiltration pathways within VC, thereby impeding water penetration. This blocking effect on infiltration channels is most pronounced in fibrous plants with vertical percolation paths, as the three-dimensional network formed by PF directly obstructs vertical root growth and promotes multi-directional expansion (Jiang et al., 2023). In the later stages of plant growth, PF exhibits signs of degradation, becoming darkened, thinned, and brittle, yet the overall network structure remains intact and functional. Over time, the interweaving and entanglement of roots and PF further reinforce the structural anchoring of VC, which also contributes to reduced infiltration. The slowing of water seepage implies prolonged water retention, representing another factor in the enhanced water holding capacity.

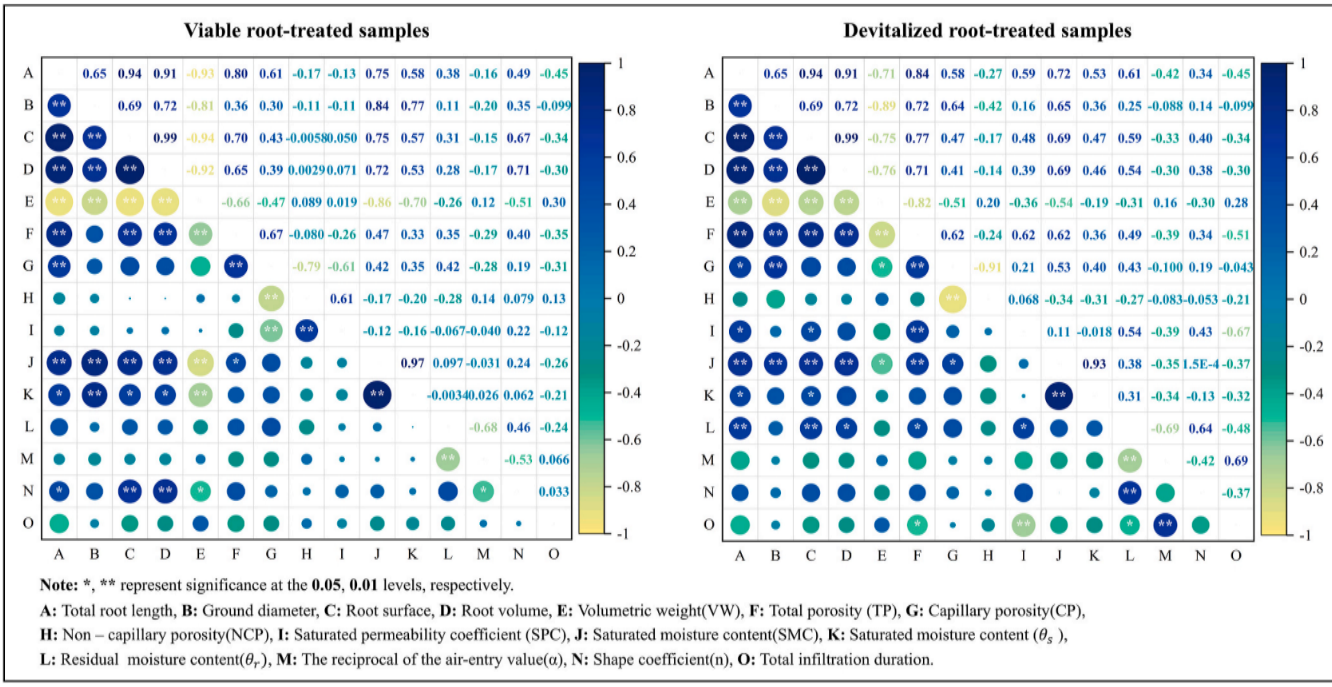

Fig. 15. Pearson Correlation Analysis of indicators for plant root traits, root-mediated VC physical properties and hydraulic characteristics.

Numerous studies have demonstrated that the incorporation of plastic fibers into soil can alter the diversity and colonization intensity of fungi such as arbuscular mycorrhizal fungi (AMF), leading to increased fungal abundance, adaptability, and root colonization, thereby promoting plant growth (Kanold et al., 2024) (Lehmann et al., 2022). We reasonably hypothesize that the incorporation of PF not only directly improves the physical structure of VC but also creates a favorable environment for the colonization and hyphal proliferation of AMF, resulting in enhanced fungal colonization intensity. As PF is a natural plant-based fiber, it exhibits stronger cementation with soil particles and is biodegradable, which may further amplify its promoting effect on mycorrhizal colonization compared to synthetic fibers. Moreover, the extensive hyphal networks of AMF can significantly expand the root absorption zone, thereby improving the efficiency of water and nutrient uptake. These combined effects synergistically promote root system development and enhance the water holding capacity of VC. The study by Zheng et al. also found that AMF significantly increases the root density of plants in soil, further stimulating root growth (Zheng et al., 2023). Concurrently, the fungi secrete more cementing substances, such as glycoproteins, which help stabilize aggregates within VC and optimize its pore structure, thereby improving its hydraulic performance (Bardgett et al., 2014). The hyphal networks themselves form stable biopores, which also influence water infiltration patterns. Du et al. reported that AMF indirectly affects soil porosity by modifying root morphological traits, resulting in a more complex soil pore system (Du et al., 2024). Within VC, the multi-material composition already results in a relatively complex pore structure. The incorporation of PF further enriches this system through dual mechanisms: modifying the physical architecture and enhancing AMF colonization intensity. This increased complexity of pore networks diversifies water infiltration pathways and attracts more extensive root and fungal colonization, ultimately establishing a positive ecological feedback cycle within the VC.

4.2. Differences and mechanisms across root systems

The inherent properties of the three plant root systems lead to fundamental differences in their basic traits. Plant G exhibits a shallow, fibrous root system, primarily distributed in the 0–4 cm shallow soil layer after 90 days, with abundant tillering. Its fibrous roots are extremely dense and feature the finest root diameter. Plant H possesses a deep taproot system, characterized by a prominent primary root that grows vertically downward, reaching 0–6 cm after 90 days. It demonstrates the greatest total root length, relatively few tillers, a medium root diameter, and the most extensive root distribution. Plant B has a deep, stoloniferous root system, also extending to 0–6 cm after 90 days, with thick and robust roots. It grows vertically in the early stage but later develops oblique or even horizontal growth, producing fibrous roots at the nodes. This species also shows high cellulose content and strong decay resistance. These differences in root attributes result in distinct spatiotemporal developmental patterns over the growth cycle. The root system of Plant G forms a high-density fibrous network in shallow soil early in growth, though its progress slows in later stages due to depth constraints. In contrast, the root system of Plant H continues to extend vertically downward throughout the growth period, with its root length advantage becoming increasingly evident over time. Plant B develops a larger root diameter early on to establish a sturdy structure, followed by a phase of explosive root length growth during later stages.

Plant roots continuously grow downward, enlarging the internal pores within the VC. Simultaneously, initial pores of varying diameters are divided or filled, reducing the compaction level of the VC. This is the same as the impact of plant roots on natural soil. Through interweaving, roots simultaneously enlarge pores while compacting soil particles in localized areas to regulate VC pore structure. By tightly enveloping and entangling soil particles, roots gradually form stable root-soil complexes. Furthermore, organic substances secreted by roots (such as polysaccharides) act as natural cementing agents, promoting soil aggregate formation and further stabilising pore structure through biochemical pathways (Kumar et al., 2017), (see Fig. 17).

From the perspective of the physical dimension interaction between PF and root systems, the dense, shallow fibrous roots of Plant G interweave with PF to form a three-dimensional network containing numerous micropores. This results in the segmentation and refinement of native macropores in VC into a multitude of interconnected micropores and capillary pores (Pulido-Moncada et al., 2022). The interweaving also creates a dense biological and physical covering layer on the surface of soil particles, significantly increasing the interface area

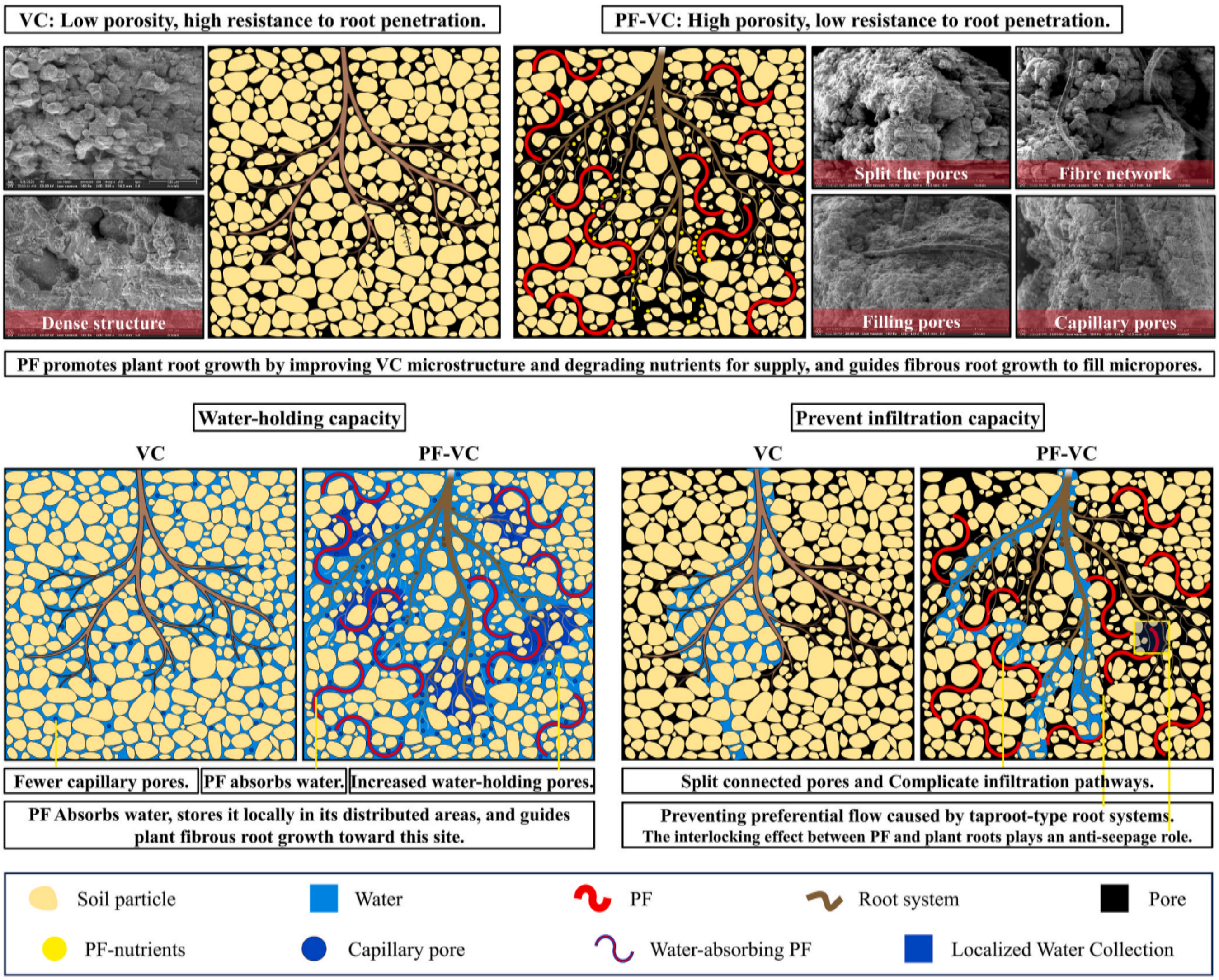

Fig. 16. Mechanism of action after PF incorporation.

between water and solid surfaces. Moreover, the extensive specific surface area and strong capillary forces considerably enhance the water holding capacity of VC. Additionally, the complex pore network increases the tortuosity of water migration paths, effectively preventing the formation of preferential flow channels and slowing down water infiltration. Plant H, with its thick and extensive root system, compacts soil particles and widens channels during downward growth, forming continuous, vertically oriented macropores. In the absence of PF, these root channels become dominant pathways for water infiltration. However, with PF incorporation, the fiber network spans across these vertical channels, functioning as a physical barrier that segments the pathways and alters flow direction, thereby increasing the complexity of seepage paths. This mechanism is reliably supported by the differential characteristics of capillary pores in VC, where PF incorporation leads to a more uniform pore size distribution. Plant B, characterized by the thickest root diameter, forms a stable mechanical framework together with PF within VC, which resists settlement of soil particles and pore changes or compaction induced by water infiltration. Furthermore, the slow decomposition rate of Plant B roots contributes to the enhanced durability of VC. Its stable and well-balanced pore structure allows VC to achieve an optimal equilibrium in hydraulic performance.

During the 90 d experimental period, in addition to the physical dimension interactions between root systems and PF, the biochemical processes resulting from their interplay also profoundly influenced the hydraulic performance of VC. Based on previous analysis, it is reasonable to postulate that PF incorporation enhances the colonization intensity of AMF and other microorganisms, synergistically creating a favorable environment for rhizosphere microbes in conjunction with plant roots. As PF and root systems interweave, fungal colonization and microbial encasement within their combined structure further promote root development. Both root growth and PF degradation processes excrete cementing organic substances, such as sugars and organic acids, while microbes concurrently produce abundant extracellular polymeric substances (EPS) (Wu et al., 2024). The colonizing fungal communities and various cementing agents adsorb onto and encase the porous, rough surfaces of the root-fiber complexes. It is noteworthy, however, that this adsorption behavior is likely root-type specific, varying across different plant species.

The extensive fibrous root surface area of Plant G leads to higher exudate secretion, resulting in the formation of hydrophilic biopolymer hydrogels at the interfaces among soil, PF, and fibrous roots. This significantly enhances the water holding capacity within its micro-pores (Wu et al., 2025). The dense fibrous roots of Plant G, together with PF, create a porous environment that provides ample interfaces for AMF colonization. The well-developed mycorrhizal network further interconnects fine pores, strengthening capillary forces and promoting uniform water distribution within the VC. Additionally, the micro-porous network formed by Plant G roots and PF offers a preferred

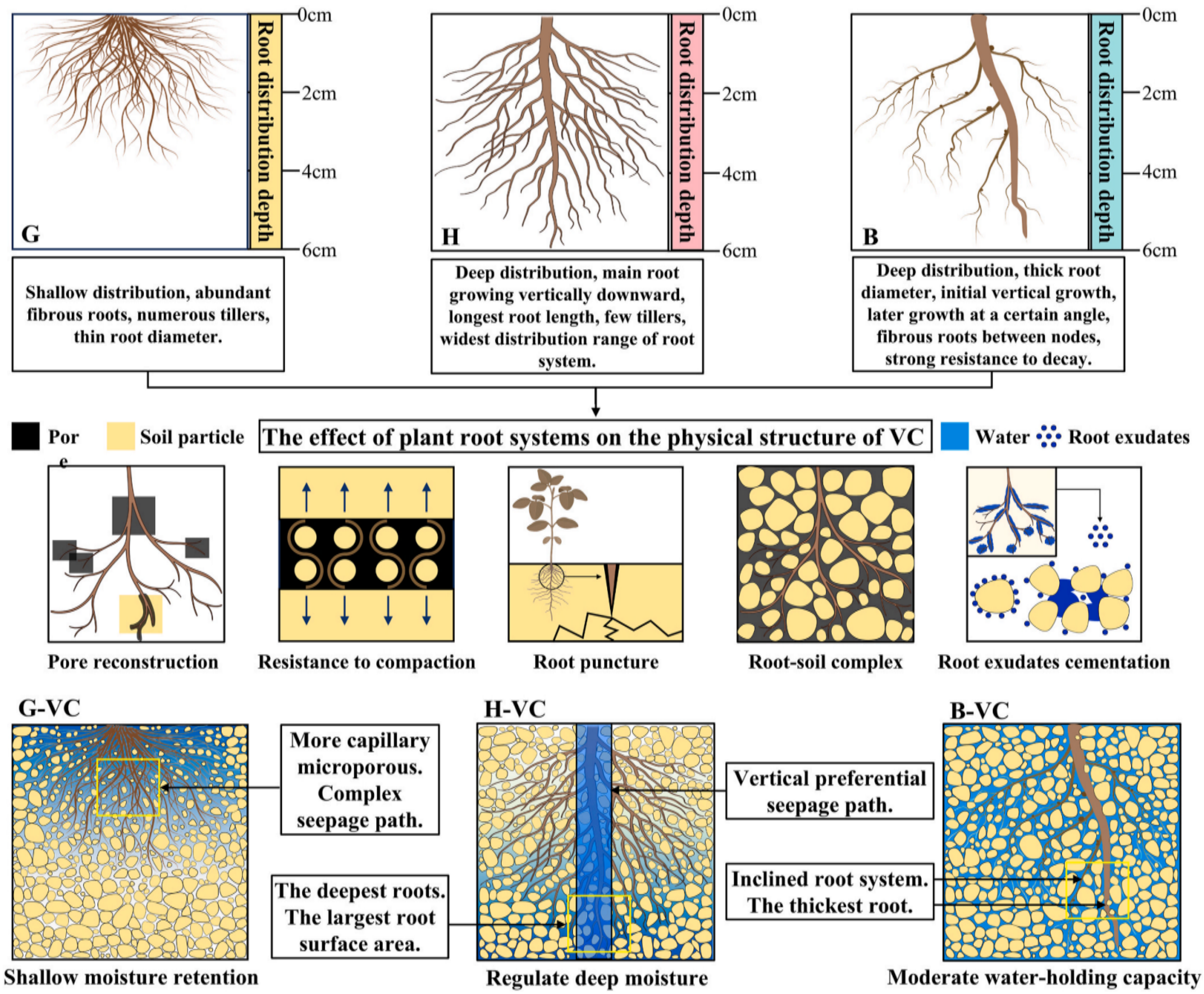

Fig. 17. Differences across root systems and synergistic/differential mechanisms of Roots-PF in regulating VC properties.

habitat for aerobic microorganisms, which produce more EPS, thereby more effectively cementing soil particles, PF, and roots into an integrated matrix (Ghezzehei et al., 2019). This constitutes a key mechanism behind the superior water holding capacity of the corresponding VC. In contrast, the taproot growth of Species H results in larger pores and weaker intertwining with PF around its main root, with root exudates and cementation primarily localized along the vertical primary root. It is known that PF may partially counteract the preferential flow effect by supporting AMF hyphal networks within the rhizosphere, thereby blocking and segmenting some of the continuous macropores formed by the large roots. However, the hyphal networks and associated exudates may be more susceptible to being transported downward by rapid water infiltration (Jia et al., 2024). Plant B exhibits balanced structural traits, with thick, slow-decomposing roots. In later growth stages, it develops fibrous roots at nodes and shifts to oblique or even horizontal growth. These characteristics facilitate the formation of a stable composite structure through physical and biochemical interactions with PF, thereby improving hydraulic performance and resulting in optimal long-term stability and decay resistance. From the perspective of unsaturated moisture transport in VC, the microstructure and surface energy of both PF and root systems are altered through microbial activity and exudate interactions. These changes influence the liquid-solid-gas contact angles and capillary behavior within the VC, ultimately shaping its overall hydraulic performance.

Overall, the interaction between plant root systems and PF is a dynamically evolving process. In the initial stage, PF acts as the primary agent modifying the physical structure of VC, promoting seed germination and early root growth. As roots develop over time, PF begins to darken and undergo gradual degradation, while the role of the roots transitions to dominance. Root growth pressure, exudates, and microbial activity subsequently reshape the framework initially established by PF. It is precisely due to the differences in growth cycles and root morphologies among plant species that become pronounced during this phase that the hydraulic performance of each VC group begins to exhibit differentiation.

4.3. Mechanisms of VC deterioration following root devitalisation

Upon root system devitalisation, their role in VC is essentially reduced from a combined biophysical mechanism to purely physical effects. The former regulates VC performance through physiological activities such as water uptake and exudate secretion, combined with physical actions like root intersperse and anchorage, whereas the latter relies solely on the morphological characteristics of the roots and their decay process (Meng et al., 2025). Living roots continuously absorb water through transpiration, maintaining the VC in an unsaturated state, thereby enabling matrix suction to contribute to structural strength and water holding capacity (Zhang et al., 2025). After root devitalisation, this moisture-regulating function is lost, leading to VC saturation and a corresponding decline in water holding capacity. Root exudates function as natural soil-binding agents within VC, working in synergy with extracellular polymeric substances from rhizosphere microbes and the hyphal networks of colonized AMF to form bio-cementation, thereby stabilising the fine water-retaining pores created by roots and PF. Following root devitalisation, this cementation effect ceases, causing the well-structured water-retaining pores to collapse or merge under external forces such as seepage, which further accelerates the reduction in VC's water holding capacity and increases its permeability. These changes have been consistently verified by the observed trends in both SMC and SWCC measurements of VC specimens (see Fig. 18).

The deterioration of VC's hydraulic performance following root devitalisation is most directly manifested in the alteration of its pore structure. Upon devitalisation, roots undergo decay and decomposition, and the remaining root channels become new, interconnected percolation pathways, fundamentally reshaping the pore network of the VC. This serves as the key mechanism responsible for the significant increase in SPC and the reduction in total infiltration time in root-devitalised specimens (Bingwei et al., 2019). However, the extent of this effect varies with root system type. In the case of Plant G, the decomposition of its abundant fibrous roots leaves behind numerous fine root channels, which still retain some water holding capacity. As a result, the decline in water holding capacity after root devitalisation is relatively moderate. Nevertheless, these numerous micropores also create localized weak points within the VC, making it more susceptible to local erosion and soil particle rearrangement. This leads to pore collapse and coalescence, eventually forming larger, unstable pores and increasing overall permeability. For Plant H, prior to root devitalisation, the three-dimensional network formed by PF provided physical interception and diversion for the continuous vertical channels created by the taproot. After the main root decomposes, it leaves behind highly continuous macropores. At this stage, the interception effect of the fibers is significantly weakened, and the remaining root channels rapidly evolve into efficient seepage pathways, resulting in the most pronounced deterioration in water holding capacity. Plant B, with its slower decomposition rate, maintains the composite framework formed with PF for an extended period even after devitalisation. Consequently, the reversal of its pore structure occurs at the slowest rate and to the mildest extent. The deterioration of its hydraulic performance is primarily attributed to the loss of biochemical cementation rather than drastic changes in pore structure. Thus, after root devitalisation, the changes in its performance indicators are the smallest and most stable among the three root types.

As root systems develop over time, their influence on VC becomes increasingly significant. Correspondingly, the loss of physiological and biological functions following root devitalisation, along with the transformation of the VC pore structure, also intensifies with time. Consequently, the deterioration of VC's hydraulic performance due to root devitalisation was relatively minor during the early experimental stage (30 days). By the middle to late stages of the experiment (60 days, 90 days), root biomass accumulation had peaked and entered an advanced decay phase. During this period, both the water holding capacity and seepage resistance of the VC deteriorated markedly (Xiaoqing et al., 2021). Additionally, although PF exhibited signs of darkening and thinning, the remaining fibers within the VC continued to function effectively. Through mechanisms such as supporting the pore structure, intercepting preferential flow, and synergistically reinforcing the soil

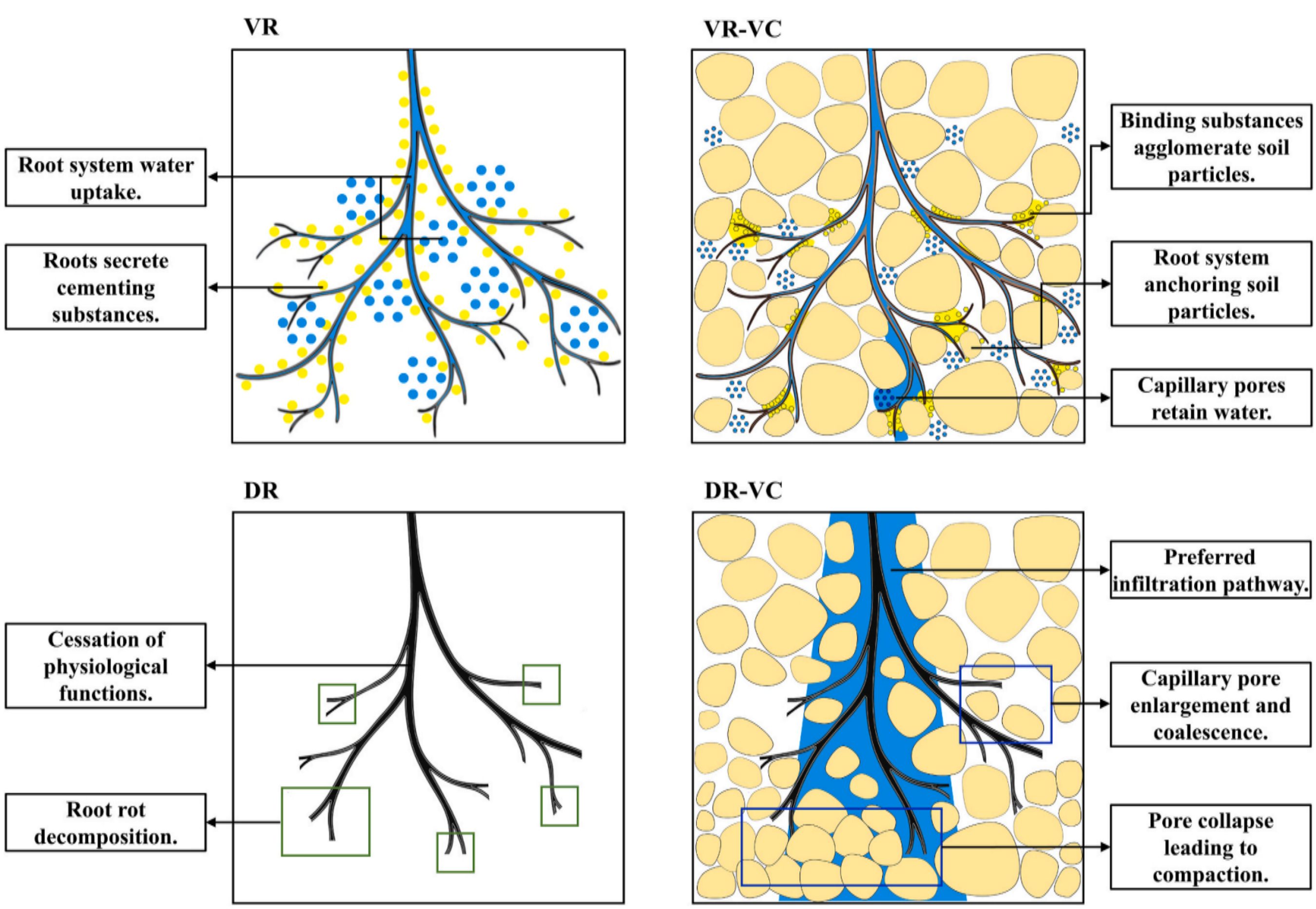

Fig. 18. Mechanisms of vegetation concrete (VC) deterioration following root devitalisation.

matrix along with roots, PF helped mitigate the decline in the hydraulic performance of VC.

4.4. Practical applications and future research directions

This study examined VC performance under controlled laboratory conditions. While this approach enabled precise measurement, it did not replicate several key environmental factors present in actual slope engineering. High-intensity rainfall events, not simulated here, could intensify preferential flow through root channels more significantly than observed. Similarly, wet-dry and freeze-thaw cycles—common in field conditions—may accelerate fibre-matrix detachment and root damage, potentially reducing long-term durability. Furthermore, natural slopes support mixed plant communities with complex root architectures, while our study used single-species systems. The hydraulic behaviour of diverse root networks requires further investigation. Finally, field applications face additional root devitalisation pressures from pests, diseases, and nutrient deficiencies that were not reproduced in our experiments. These factors could substantially alter hydrological performance and merit dedicated study (Dimitriadis et al., 2024c) (Liu et al., 2025c).

This study reveals fundamental fibre-root-substrate interaction mechanisms that are widely applicable beyond expansive soil slopes. PF and root development regulate VC hydraulic performance, providing a transferable framework for various ecological engineering applications. These include red soil slopes, loess slopes, steep rocky slopes, roadway slopes, and river embankment slopes, among others. Furthermore, understanding pore structure modification provides key insights for designing VC systems for different climates—from water-limited regions prioritising retention to high-rainfall areas requiring drainage. Regarding long-term performance, our findings provide important insights for sustainable slope protection. Although focusing on the early establishment phase, this period represents the most critical window for VC performance, when the substrate is most vulnerable before roots fully develop. We observed that palm fibre maintained synergistic function with roots throughout the experiment, indicating its lasting role in preserving hydraulic performance. However, vertical taproot-dominated systems maintained preferential flow pathways, revealing permeability issues that may compromise long-term stability. These findings emphasise the importance of selecting suitable plant species and incorporating fibre reinforcement to ensure both immediate and long-term performance of ecological slope protection systems.

This study clarifies how PF and root architecture affect VC hydraulic performance, though all experimental designs have limitations. First, pot experiments restricted deep root growth and poorly represented hydrological processes at larger scales, differing from actual slope responses. Although we used replicate samples, distributing them across treatment groups reduced sample sizes per group. This may have weakened our ability to detect subtle interactions. Second, the laboratory setting excluded complex field conditions. However, at field sites, factors like wet-dry cycles, freeze-thaw cycles, and variable rainfall continuously affect VC composite structure. Our short-term study missed these cumulative effects. Third, fibre geometry and surface properties vary among types, potentially leading to different interactions with plant roots. Similarly, plant responses to different environments may produce root architectures and hydraulic effects unlike ours. We also lacked biochemical analyses of how PF promotes root growth and modifies VC structure and hydraulic performance. Based on these limitations, future work should monitor VC hydraulic performance longer-term under actual slope conditions. Studies should also test more fibre types and plant communities to find common regulatory mechanisms. Advanced techniques like CT should quantify pore structures non-destructively. Measuring microbial activity, root exudates, and fungal colonization in PF-amended VC would further verify the mechanisms.

5. Conclusions

Research on plant root hydraulics is extensive in natural soils, but limited in engineered materials like vegetation concrete (VC). For VC, studies often focus on mechanical properties, not hydraulic performance. The combined effect of roots and common additives like palm fibre (PF) is also rarely studied in a dynamic context that considers different root types and vitality changes over time. To address these knowledge gaps, this study investigated how PF and three herb species with different root architectures influenced VC hydraulics across growth periods (30, 60, 90 days) and root vitality states (viable roots, VR; devitalised roots, DR). We analysed root morphology, VC physical properties, soil water characteristics, permeability, and soil-column infiltration. Correlation analysis revealed how fibre-root-substrate interactions regulate VC hydraulic performance. The main conclusions are as follows:

1. PF incorporation reconstructed the pore structure of VC by forming a three-dimensional network through dividing and filling connected pores. This comprehensively reduced VC bulk density, increased VC porosity, and created a low-resistance growth environment for root development. PF incorporation promoted plant root development, while the physical intertwining of PF with different plant roots formed distinct composite structures: forming micropore networks with the fibrous roots of *Cynodon dactylon*; intercepting and segmenting preferential flow channels formed by the vertical taproots of *Lolium perenne*; and constructing stable skeletal structures with the coarse roots of *Paspalum notatum*.
2. Different root architectures induced distinct hydraulic responses in VC. Fibrous roots significantly increased root-soil contact interfaces through their dense networks and, synergistically with PF, created abundant capillary pores that enhanced localized water retention. Vertical taproots formed deep preferential flow channels, though PF networks effectively mitigated the associated seepage risks. Coarse roots combined with PF to construct stable composite skeletons, achieving optimal balance between water retention and structural stability.
3. A notable physical-biochemical coupling effect characterises fibre-root interactions. Beyond providing physical scaffolding, PF) improved the microhabitat to enhance microbial activity. Root exudates and microbial extracellular polymers acted as biological cementing agents, further stabilising VC pore structures. This synergistic effect proved most pronounced in fibrous root-PF complexes, establishing a sustained enhancement mechanism for VC's water holding capacity.
4. Root devitalisation triggered a cascade of deterioration in VC. The cessation of active water uptake by devitalised roots compromised water holding capacity, while depletion of biological cementing agents undermined pore stability. Simultaneously, decomposing root channels established new seepage networks that altered hydraulic pathways. This degradation process exhibited strong root-type dependency.
5. VC exhibited clear spatiotemporal patterns in its hydraulic performance. Over time, hydraulic regulation progressively shifted from fibre dominance to root control as plants matured. Spatially, a distinct hydraulic stratification emerged, mirroring the vertical distribution of roots. Upper soil layers, enriched with roots, demonstrated enhanced water retention and slower seepage rates. Conversely, deeper zones with sparse root systems displayed higher bulk density, lower porosity, and consequently faster water transmission—a process increasingly governed by the intrinsic properties of the VC matrix itself. Within these deeper layers, vertical roots further intensified preferential flow pathways.

Based on these findings, we recommend tailored fibre-plant combinations for VC with specific hydraulic performance requirements. For shallow anti-permeability and soil-water conservation, PF with *Cynodon dactylon* is optimal. For long-term structural stability and degradation resistance, PF with *Paspalum notatum* is preferred. *Lolium perenne* alone should be avoided in seepage-sensitive slope applications; instead, it should be combined with PF and fibrous-root species to mitigate preferential flow risks. From a root vitality perspective, slope vegetation should be treated as living infrastructure requiring regular monitoring and maintenance. When significant succession or degradation occurs in vegetation communities, timely assessment and replanting are necessary to preserve the hydrological and ecological functions of the vegetation. Collectively, the mechanisms of fibre-root synergy and antagonism revealed in this study provide both a predictive framework for understanding VC hydrological behaviour in field conditions and a scientific basis for developing more resilient, sustainable, and cost-effective ecological slope protection strategies.

CRediT authorship contribution statement

Xu Yang: Writing – original draft, Visualization, Investigation, Data curation. **Jian Chen:** Validation, Methodology, Formal analysis, Conceptualization. **Wenhao Dong:** Supervision, Project administration, Data curation. **Liming Liu:** Writing – review & editing, Resources, Funding acquisition. **Bin Wu:** Visualization, Software, Methodology. **Dong Xia:** Validation, Conceptualization. **Daxiang Liu:** Writing – review & editing, Funding acquisition. **Bingqin Zhao:** Software, Project administration. **Yang Xu:** Data curation. **Mingyi Li:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2025.128360.

Data availability

Data will be made available on request.

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